

Project 2 Research Stage — transcript

- **Project:** ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator
- **Repository:** <https://github.com/rg-sureshkumar-goat/AI-Footprint-Calculator>
- **Branch:** project-2 (HEAD at start of session: d871894; at save: d871894)
- **Date:** 2026-09-26
- **Learner:** RG Sureshkumar
- **Agent:** Claude Code (Opus 5.5)
- **Stage command:** start research (instructions in research.md, Commands section)
- **Outcome:** Research recorded in research.md (11 audience-need sections, 12 possible features, 58 source assessments); five features selected and explicitly approved by the learner.

Note on redaction. Sanitised following transcripts/README.md: the author's email address, macOS home path and username, and temporary/session paths are replaced with placeholders. No messages were removed or reworded.

Note on tool calls. Each tool call is shown as a one-line summary in square brackets, followed by its output truncated to 400 characters. The agent's internal reasoning is not included. The complete local session log is kept locally and not committed.

Note on the final turn. The last learner turn (save transcript) is followed by the tool calls that generated this file; the agent's closing confirmation was written after the file was generated and so is not included.

1. Learner (RG Sureshkumar)

start research

Agent

[Tool: Bash — Read research instructions, brief, and file list]

```
Exit code 1
# Research

> EDITING DIRECTIVE: USER AND AGENT EDIT THIS FILE COLLABORATIVELY. THE USER MUST REVIEW AND
APPROVE ITS CONTENT.

Purpose of this file: Develop and record the evidence and decisions that will guide the
technical specification.

## Instructions for the user

You are responsible for the ethics, accuracy, and fairness of the research. Direct the inquiry
toward useful question ... [truncated]
```

[Tool: Bash — Read brief and list project files]

```
# Project 2 Brief
```

```
> EDITING DIRECTIVE: DO NOT EDIT THIS FILE. IT IS THE AUTHORITATIVE PROJECT FRAME.
```

```
Roles in the project documents: The user is the person directing and working with the AI agent. The agent supports the user's work but does not make or approve decisions on the user's behalf. Employees are the intended audience for the completed calculator, and the named reference profiles are fi ... [truncated]
```

Research stage is starting. Here's where things stand.

What this stage produces. A reviewed draft of [research.md](#) with four sections, done in order: 1. **Audience needs.** What Alex, Jordan and Robin do and need, with evidence kept apart from assumptions we still have to check. 2. **Possible features.** Several short ideas before any are chosen. 3. **Source assessments.** Each source gets a full citation, a link I've checked, the exact claim we'd use, its limits, and a use/qualify/reject call. 4. **Selected features.** Five features, which **only you can approve**.

Limits from the brief that will shape our choices: - At least **2 features** must improve how professional AI use is represented. - At least **2** must connect AI use to broader digital life. - The **5th** must meet the strongest need left after research. - Together they must serve all three profiles. - The calculator must not argue that AI is harmless or that it's harmful.

How I'll work. One step at a time. I'll check every source directly before citing it, look for evidence that cuts the other way, and say plainly when I can't verify something. I'll draft text for you to review, and I won't mark anything approved.

Starting point: section 1, Audience needs. The profiles point to several evidence questions, including: - Energy and water for image and video generation (Alex) - Coding-agent and long-session use (Jordan) - Video calls, streaming, gaming and social media footprints (all three) - What makes environmental estimates trustworthy to skeptical readers (Robin)

One question to start: Should we go through the needs profile by profile (Alex, then Jordan, then Robin), or activity by activity (professional AI use first, then everyday digital habits), since several activities are shared across profiles? Tell me which you prefer, and whether you already have sources or concerns you want looked at first.

2. Learner (RG Sureshkumar)

lets go activity by activity

Agent

[Tool: Bash — Inspect existing calculator data and README]

```
/**
 * data.js – reference figures for the calculator.
 *
 * Per-model energy / carbon / water estimates (mean plus EcoLogits' own 95%
 * interval) come from the EcoLogits library v0.10, the engine behind the
 * EcoLogits calculator at huggingface.co/spaces/genai-impact/ecologits-calculator.
 * Every other figure is sourced in the methodology section of index.html.
 *
 * Carried over unchanged fro ... [truncated]
```

Going activity by activity. First I checked what the calculator already covers, so we research the gaps rather than repeat Project 1.

What it covers now ([assets/js/data.js](#)): - **Text AI only**. Per-prompt energy, carbon and water from EcoLogits, by model and output length. The longest option is a single ~100k-token "coding / agent session." - **Comparisons**. Everyday items like coffee, showers and flights, plus an hour on a PS5. There is no way to enter streaming, video calls, social media or other device use.

Proposed activity order. The profile tags show who each activity mainly concerns. They are starting assumptions, not findings.

Professional AI use 1. **Image generation**: Alex 2. **Video generation**: Alex 3. **Coding agents and long, irregular AI sessions**: Jordan. This checks whether one fixed "session" size is realistic. 4. **Occasional text AI**: Robin. This is already covered, so mainly a check of whether current figures still hold up.

Broader digital life

5. **Video calls**: Jordan, Robin
6. **Video and music streaming** on phone, laptop and TV: Alex, Robin
7. **Social media use**: Alex, Robin
8. **Gaming**, desktop PC and console: Jordan
9. **Device manufacturing (embodied impacts)**: all three. This covers phones, laptops, TVs and PCs.

Across all activities

10. **Water**: Alex. How water is counted (on-site cooling vs. electricity generation) and where, for example in Los Angeles.
11. **Trust and communication**: Robin. What evidence says about presenting uncertainty, sources and company-sponsored tools to skeptical readers.

For each activity I'll find sources, check them directly, look for evidence that disagrees, and draft short notes for **Audience needs**. Each note will separate *evidence* from *assumptions still to check*. You review before anything goes into [research.md](#).

Question: Does this list and order work for you? Tell me what to add, drop or reorder, or say go and I'll start with image generation.

3. Learner (RG Sureshkumar)

yes this list works, lets start with image generation

Agent

[Tool: ToolSearch — select:WebSearch,WebFetch]

[Tool: WebFetch — https://arxiv.org/abs/2311.16863]

```
# Citation and Key Findings

**Full Citation:**
Luccioni, Alexandra Sasha; Jernite, Yacine; and Strubell, Emma. "Power Hungry Processing:
Watts Driving the Cost of AI Deployment?" *ACM Conference on Fairness, Accountability, and
Transparency (ACM FAccT '24)*, June 3–6, 2024, Rio de Janeiro, Brazil.

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## Energy Consumption Data

Unfortunately, the webpage content provided does not include the s ... [truncated]
```

[Tool: WebSearch — query: MIT Technology Review "We did the math on AI's energy footprint" image generation joules Stable Diffusion]

```
Web search results for query: "MIT Technology Review "We did the math on AI's energy
footprint" image generation joules Stable Diffusion"

Links: [{"title":"Toward Sustainable Generative AI: A Scoping Review of Carbon Footprint and
Environmental Impacts Across Training and Inference
Stages","url":"https://arxiv.org/pdf/2511.17179"}, {"title":"We did the math on AI’s energy
footprint. Here’s the sto ... [truncated]
```

[Tool: WebSearch — query: AI Energy Score Hugging Face text-to-image energy per 1000 queries leaderboard]

```
Web search results for query: "AI Energy Score Hugging Face text-to-image energy per 1000
queries leaderboard"

Links: [{"title":"How output length shapes AI’s energy footprint / AI
World","url":"https://aiworld.eu/story/how-output-length-shapes-ais-energy-footprint-"},
{"title":"Hugging Face presents AI Energy Score | by Sulbha Jain |
Medium","url":"https://medium.com/@sulbhajain/hugging-face-pres ... [truncated]
```

[Tool: WebFetch — https://arxiv.org/html/2311.16863]

```
# Image Generation Energy Figures from Research Paper

## Energy Consumption Data
```

****Mean energy per 1,000 inferences:**** The study reports image generation consumed "2.907" kWh on average, with standard deviation of 3.31 kWh across tested models.

****Most energy-intensive model:**** The stable-diffusion-xl-base-1.0 model generated "1,594 grams of CO₂eq for 1,000 inferences," representing the highest ... [truncated]

[Tool: WebFetch — <https://www.technologyreview.com/2025/05/20/1116327/ai-energy-usage-climate-footprint-big-tech/>]

AI Image Generation Energy Figures

****Article:**** "We did the math on AI's energy footprint. Here's the story you haven't heard"

****Authors:**** James O'Donnell and Casey Crownhart

****Date:**** May 20, 2025

****Source:**** MIT Technology Review

Stable Diffusion 3 Medium Measurements

For standard-quality images (1024 x 1024 pixels), the researchers found:

– ****Base energy:**** "about 1,141 joules of GPU ... [truncated]

[Tool: WebFetch — <https://huggingface.github.io/AIEnergyScore/>]

AI Energy Score Methodology

Energy Measured

The AI Energy Score measures ****GPU energy consumption only****, specifically on NVIDIA H100 80GB GPUs. While the system tracks total energy across CPU, RAM, and other components for transparency, only GPU energy (measured in watt-hours) determines the official ratings.

As stated in the documentation: "only GPU energy consumption is used for scoring ... [truncated]

[Tool: Bash — Download Luccioni paper and extract image-generation passages]

... classification requiring less energy (median of 0.0068 kWh for 1,000 inferences) than image generation (1.35 kWh) and, conversely, text generation (0.042 kWh) requiring more than text classification (0.0023 kWh). For comparison, charging the average smartphone requires 0.022 kWh of energy (US Environmental Protection Agency, 2024), which means that the most efficient text generation model ... [truncated]

[Tool: WebSearch — query: "The Hidden Cost of an Image" energy consumption AI image generation arXiv 2025]

Web search results for query: ""The Hidden Cost of an Image" energy consumption AI image generation arXiv 2025"

Links: [{"title": "[2506.17016v1] The Hidden Cost of an Image: Quantifying the Energy Consumption of AI Image Generation", "url": "https://arxiv.org/abs/2506.17016v1"}, {"title": "

[2506.17016] The Hidden Cost of an Image: Quantifying the Energy Consumption of AI Image Generation", "url": "http ... [truncated]

[Tool: WebSearch — query: OpenAI OR Google OR Midjourney disclosed energy per image generation Wh 2026]

Web search results for query: "OpenAI OR Google OR Midjourney disclosed energy per image generation Wh 2026"

Links: [{"title": "Best AI Image Generators Compared: Pricing and Use Cases (2026) | AI Weekly", "url": "https://aiweekly.co/learning-ai/ai-applications/best-ai-image-generators"}, {"title": "Midjourney", "url": "https://en.wikipedia.org/wiki/Midjourney"}, {"title": "Best AI Image Generation Models ... [truncated]

[Tool: WebFetch — https://www.technologyreview.com/2025/05/20/1116331/ai-energy-demand-methodology/]

Image Generation Methodology Summary

Model & Hardware:

"Stable Diffusion 3 from Stability AI" was tested, specifically "one of the most popular sizes of Stable Diffusion 3, with 2 billion parameters." Testing used "the same hardware as our text and image tests"—an Nvidia H100 GPU.

Steps:

"For a given step count and model, all images generated have the same energy footprint." The author ... [truncated]

[Tool: WebFetch — https://arxiv.org/html/2506.17016]

Energy Consumption of AI Image Generation – Key Findings

Exact Energy Figures

Lowest energy model: LCM SSD_1B at 8.6×10^{-5} kWh (0.31 J)

Highest energy model: Lumina at 4.08×10^{-3} kWh (14.7 J)

Difference: 46x

These measurements represent "the sum of CPU, GPU, and RAM energy consumption" across the entire system.

Testing Hardware & Conditions

– **GPU:** NVIDIA GeForce ... [truncated]

[Tool: Bash — Download Bertazzini paper and extract energy figures]

... rics to study potential trade-offs between energy consumption and generated image quality. Results show that image generation models vary drastically in terms of the energy they consume, with up to a 46x difference. Image resolution affects energy consumption inconsistently, ranging from a 1.3x to 4.7x increase when doubling resolution. U-Net-based models tend to consume less than Transformer- ... [truncated]

[Tool: WebFetch — <https://ecologits.ai/latest/>]

```
# EcoLogits Supported Modalities

EcoLogits now supports video generation in addition to text generation. The homepage
announces:

> "🎬 EcoLogits now estimates environmental impacts for AI-generated videos!"

The library tracks environmental impacts for generative AI models at inference across multiple
modalities. The documentation includes sections on "Video Generation" methodology and
tutorials, ... [truncated]
```

[Tool: Bash — Search EcoLogits homepage for image and video support]

```
... e+"."+e,JSON.stringify(_))}catch(e){}} Skip to content 🎬 EcoLogits now estimates
environmental impacts for AI-generated videos! Read the announcement . var
el=document.querySelector("[data-md-component=announce]");if(el){var content=el.querySele
... _ecologits log model_repository exceptions electricity_mix_repository status_messages
impacts impacts dag llm modeling video estimations estimati ... [truncated]
```

Image generation notes are drafted below. I checked every figure against the source text itself, not a search-result summary. That mattered: one automated summary converted a unit wrongly (it gave 0.31 J where the paper means about 310 J), and I've used the paper's own numbers.

Draft: Image generation (serves Alex)

Evidence (checked directly)

Source	Setup	Energy per image
Luccioni, Jernite & Strubell, "Power Hungry Processing," <i>ACM FAccT '24</i> (arXiv)	Open models from 2022–23, 1,000 images each, one NVIDIA A100 GPU (idle power of the other GPUs in the machine included)	Median 1.35 Wh , mean 2.9 Wh , highest model (SDXL) 11.5 Wh
O'Donnell & Crownhart, <i>MIT Technology Review</i> , 20 May 2025 (article , methodology); measured by the ML.Energy team	Stable Diffusion 3 Medium, 1024×1024, H100 GPU	0.32 Wh GPU only at 25 steps; ~0.63 Wh after doubling for cooling and other overhead; ~1.2 Wh at 50 steps
Bertazzini et al., "The Hidden Cost of an Image," arXiv preprint, June 2025 (arXiv)	17 open models, consumer RTX 4090, whole system (CPU + GPU + RAM)	Median 0.086 Wh (LCM_SSD_1B) to 4.08 Wh (Lumina), a 46× spread

Other findings: - **Settings change the result a lot.** - The choice of model alone can change energy up to 46× (Bertazzini). - Doubling resolution raised energy 1.3× to 4.7×, depending on the model. - Doubling steps roughly doubled energy (MIT TR). - Prompt length had no statistically significant effect (Bertazzini). - **Higher quality doesn't always cost more.** Some of the highest-quality models were also among the most efficient (Bertazzini). - **For comparison,** the calculator's current EcoLogits figure for one GPT-5.5 chatbot reply is ~2.7 Wh ([data.js](#)). So one image falls roughly in the same range as a text reply, from well below it to a few times above it, depending on the model.

Limitations and contrary points

- **None of these sources measured the commercial tools creative staff are likely to use**, such as Midjourney, OpenAI's image models, Google's or Adobe Firefly. I found no per-image energy disclosure from any of them. Bertazzini left DALL·E and Midjourney out because they aren't publicly accessible, and MIT TR describes the closed providers as a "black box." That is a search result, not proof that no disclosure exists.
- **The hardware differs across studies:** data-center A100 and H100 GPUs versus a consumer RTX 4090. Measurement scope differs too (GPU only vs. whole system), so the numbers aren't directly comparable.
- **MIT TR's doubling for overhead is a rule of thumb** that its own methodology says is untested for diffusion models. It's also unclear whether the ~1.2 Wh figure at 50 steps is GPU-only or already doubled.
- **The models measured are older open-source ones.** Newer commercial models may use different architectures, so figures could be higher or lower.
- **Bertazzini is a preprint.** I haven't found a peer-reviewed version.
- **EcoLogits, the calculator's current data source, covers text and video but not images** (ecologits.ai). Image figures would need a new source.

Assumptions still to check

- **Water per image:** I found no direct figure. Any water number would have to be derived from energy × data-center and grid water intensity, which is an assumption to state openly.
- **How many images a professional generates per task or day**, including discarded variations. I have no evidence on this yet, and it may dominate the total more than the per-image figure does.
- **Which tools the company's staff actually use.** That's for you or the client to confirm.

Question: Should I add this to the **Audience needs** and **Source assessments** sections of research.md as a draft now, or do you want to change or challenge anything first?

4. Learner (RG Sureshkumar)

yes, add it to research.md

Agent

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

I've added the image generation notes to [research.md](#), marked as a draft for your review.

- **Audience needs:** a new "1. Image generation (Alex)" section with evidence, limitations, and assumptions still to check.
- **Source assessments:** entries S1 to S3 with the full citation, link, figures used, what I checked, and limitations.

The confidence levels and use decisions are proposals, not decisions. I proposed "moderate; use with qualifications" for all three sources. Please confirm or change them. Nothing is marked approved.

Next on the list is **2. Video generation (Alex)**. EcoLogits, the calculator's current data source, now covers video, so I'll check it first, along with MIT TR's video figure and any independent measurements that support or contradict it.

Question: Should I go ahead with video generation, or do you want to review or change the image section first?

5. Learner (RG Sureshkumar)

lets move on to video generation

Agent

[Tool: Bash — Read EcoLogits video generation methodology page]

(Bash completed with no output)

[Tool: WebSearch — query: "Video Killed the Energy Budget" text-to-video energy consumption arXiv]

Web search results for query: ""Video Killed the Energy Budget" text-to-video energy consumption arXiv"

Links: [{"title": "[2509.19222] Video Killed the Energy Budget: Characterizing the Latency and Power Regimes of Open Text-to-Video Models", "url": "https://arxiv.org/abs/2509.19222"}, {"title": "Video Killed the Energy Budget: Characterizing the Latency and Power Regimes of", "url": "https://arxiv.org ... [truncated]"}]

[Tool: Bash — Extract video energy figures and find EcoLogits video docs link]

```
href="/blog/2026/06/18/video-impacts/"
href="methodology/video_generation/"
href="reference/estimations/video/"
href="reference/impacts/video/"
href="tutorial/video_generation/"
... uncement-text { color: var(--arxiv-ink, #1c1a17); } Report GitHub Issue × Title: Content
selection saved. Describe the issue below: Description: Submit without GitHub Submit in GitHub
arXiv is now an independent nonpro ... [truncated]
```

[Tool: Bash — Extract cross-model comparison and limitations text]

5.2 Cross-Model Comparison 6 Discussion Implications for efficiency. Validated linear regime in steps. Opportunities for model-level improvements. Broader implications. 7 Limitations and Conclusion Limitations. Conclusion. References A Detailed FLOP Derivations and Scaling Laws Conventions. A.1 Latent Tokenization and Shapes A.2 Self-Attention in the DiT A.3 Cross-Attention (Video → \to Text) A.4 ... [truncated]

[Tool: Bash — Extract body of cross-model comparison section]

5.2 Cross-Model Comparison Finally, we compare average GPU energy consumption, latency, and component-wise energy shares across seven open-source text-to-video models under their default generation settings (Figure 5). (a) GPU energy per model for one video (b) Generation latency per model for one video (c) Energy shares (GPU/CPU/RAM) Figure 5 : Cross-model comparison of energy and latency. Top: ... [truncated]

[Tool: Bash — Read EcoLogits video methodology and announcement pages]

Exit code 1

HTTP 200
12382

API Reference _ecologits log model_repository exceptions electricity_mix_repository status_messages impacts impacts dag llm modeling video estimations estimations video utils utils opentelemetry range_value tracers tracers mistralai_tracer google_genai_tracer huggingface_tracer anthropic_tracer openai_tracer utils cohere_tracer litellm_tracer Blog Blog Blog Archive Arc ... [truncated]

[Tool: WebSearch — query: Jegham 2026 video generation energy Sora Veo latency regression environmental impact arXiv]

Web search results for query: "Jegham 2026 video generation energy Sora Veo latency regression environmental impact arXiv"

Links: [{"title":"2026 - EcoLogits","url":"https://ecologits.ai/latest/blog/archive/2026/"}], [{"title":"Blog - EcoLogits","url":"https://ecologits.ai/latest/blog/"}], [{"title":"Lights, Camera, Carbon: Architectural Scaling Laws for Video Generation Energy Consumption","url":"http ... [truncated]

[Tool: Bash — Read EcoLogits video announcement and MIT TR video passages]

Back to index Samuel Rincé Founder & Maintainer Metadata June 18, 2026 in Methodology 3 min read AI Videos in EcoLogits We are happy to release our first methodology for assessing the environmental footprint of AI-generated videos . This is a major step forward in modeling new GenAI use cases beyond text generation, and this contribution is part of the work pursued with the GenAI footprint Allianc ... [truncated]

[Tool: Bash — Extract Jegham et al. 2026 abstract and proprietary model estimates]

9118 jeg.txt
64619

Abstract Download PDF Abstract 1 Introduction 2 Background Notations 2.1 Architectural Decomposition 2.2 Compute-bound Nature of Diffusion Models 3 Scientific Approach 3.1 Theoretical Foundation 3.2 Experimental Methodology 4 Results 4.1 The Compute-Bound Nature of Diffusion Models 4.2 Architectural Decomposition & Forward Validation 4.2.1 The Linear & Quadratic Decomposition of ... [truncated]

[Tool: Bash — Extract Jegham et al. discussion and limitations]

5 Discussion and Limitations This study presents a framework to empirically characterize the energy required by T2V models across a range of architectures, obtained via prompting and direct energy measurement and extended to proprietary systems via API-derived generation times. This contributes to a more rigorous empirical and architectural understanding of a modality whose footprint continues to ... [truncated]

Video generation notes are drafted below. All figures come from the source text itself. One thing matters for Robin: EcoLogits' video method was funded by an alliance led by Publicis Groupe, an advertising company. I've recorded that neutrally below.

Draft: Video generation (serves Alex)

Evidence (checked directly)

Source	Setup	Energy per video
Delavande, Pierrard & Luccioni, "Video Killed the Energy Budget," arXiv, Sept 2025 (arXiv); Hugging Face; video-generation workshop paper	7 open models at their default settings, one H100 GPU	0.14 Wh (AnimateDiff) to over 415 Wh (WAN2.1-14B), about a 3,000× spread. CogVideoX-5B: ~25 Wh total
Jegham, Gamazaychikov & Luccioni, "Lights, Camera, Carbon," arXiv, July 2026 (arXiv)	Measured 6 open models on 3 GPU setups, then <i>estimated</i> commercial models from API generation times plus assumed hardware	One 8-second 720p video: Veo 3 ~20–43 Wh , Seedance-1 ~72–88 Wh, Runway Gen-4.5 ~242–411 Wh, Sora 2 Pro ~315–534 Wh . Sora 2 Pro 12-second 1080p: ~1,313 Wh (mean)
O'Donnell & Crownhart, <i>MIT Technology Review</i> , 2025 (same article as S2)	CogVideoX, measured by Luccioni with CodeCarbon	Older low-quality model ~30 Wh (109,000 J); newer 5-second, 16 fps model ~940 Wh (3.4 MJ)
EcoLogits video method, June 2026 (method , announcement)	Built on Jegham et al.: latency regressions × whole-server power × data-center overhead (PUE)	Gives energy, carbon, water and hardware-manufacturing ranges per video for commercial providers including Google, OpenAI, Runway, Kling and ByteDance

What drives the cost: - Energy rises roughly with the *square* of resolution and of video length, and in direct proportion to the number of denoising steps (Delavande). - So doubling the length or the resolution costs much more than double.

For scale, using the calculator's figure of ~2.7 Wh for one GPT-5.5 chatbot reply: - One Veo 3 clip ≈ about 10 replies. - One Sora 2 Pro 1080p clip ≈ several hundred replies. - Unlike images, video can be far above text.

Limitations and contrary points

- **The commercial-model figures are estimates, not measurements.** Jegham et al. inferred them from API timings and *assumed* hardware, for example that Veo runs on Google TPU v6e chips. The providers haven't confirmed them.
- **The sources disagree for the same model family.** CogVideoX-5B is ~25 Wh in Delavande, but MIT TR reports ~940 Wh for a newer CogVideoX. That is probably due to different versions and settings, but I can't fully reconcile the two.
- **EcoLogits assigns the whole server's power to each request**, with no sharing across simultaneous requests. That likely pushes estimates up.
- **Two of the key sources are preprints** (Jegham, Delavande), and all involve the same researcher (Luccioni). They are not independent confirmations of each other.
- **Funding:** EcoLogits' video work came from the "GenAI footprint Alliance," a Publicis Groupe initiative whose members include AXA and Engie. That doesn't make it wrong, but it's worth telling readers who worry about corporate motives.
- **Audio isn't well covered.** Delavande didn't study it, and Jegham calls their audio estimates less grounded.

Assumptions still to check

- **How many clips a professional generates per finished shot.** Jegham et al. cite reports that one AI-generated Coca-Cola ad involved over 70,000 generated videos. That is secondhand from a news report, one extreme case, and not yet verified by me. If iteration dominates the total, a calculator should probably ask about attempts, not just final outputs.
- **Water per video:** EcoLogits' figure is *derived* (energy × water intensity), not measured.
- **Which video tools the company uses**, and at what length and resolution.
- **Whether EcoLogits' calculator output matches the paper's figures.** I haven't run it yet. That's a check for the specification stage.

I'd propose all four sources as "**use with qualifications**," with Jegham/EcoLogits as the main data path because they cover commercial tools. That's your decision.

Question: Should I add this to [research.md](#) the same way as the image section (Audience needs plus Source assessments S4 to S6, with MIT TR folded into S2), or do you want changes first?

6. Learner (RG Sureshkumar)

yes add it to research.md

Agent

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

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The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

I've added the video generation notes to [research.md](#) as a draft.

- **Audience needs:** a new "2. Video generation (Alex)" section with evidence, limitations, and assumptions still to check.
- **Source assessments:**
 - S2 (MIT TR) now also includes its CogVideoX video figures. I noted that they conflict with S4.
 - S4 (Delavande et al. 2025), S5 (Jegham et al. 2026) and S6 (EcoLogits video method) are new entries.

Everything is still a proposal for you to confirm. I proposed "use with qualifications" for all three new sources. For S5, I rated the estimates for commercial models "moderate to low," to be shown as ranges and labelled as estimates. Nothing is marked approved.

Two items need your judgement at some point: 1. The unresolved conflict between the CogVideoX figures (~25 Wh vs. ~940 Wh). 2. Whether the Publicis Groupe funding disclosure belongs in the calculator itself or only in the research notes.

Next on the list is **3. Coding agents and long, irregular AI sessions (Jordan)**. The calculator currently uses one fixed ~100,000-token "agent session" from EcoLogits. I'd look for evidence on how many tokens real coding-agent sessions actually use and how much they vary, and whether per-token energy figures hold up for long contexts.

Question: Should I go ahead with coding agents, or deal with either of those two items first?

7. Learner (RG Sureshkumar)

lets move on to coding agents

Agent

[Tool: WebSearch — query: Simon Couch electricity use of AI coding agents Claude Code session Wh]

Web search results for query: "Simon Couch electricity use of AI coding agents Claude Code session Wh"

Links: [{"title": "Electricity use of AI coding agents | Simon P. Couch — Simon P.

```
Couch", "url": "https://simonpcouch.com/blog/2026-01-20-cc-impact/"), {"title": "Electricity use of AI coding agents", "url": "https://simonwillison.net/2026/Jan/20/electricity-use-of-ai-coding-agents/"), {"title": "my ai ... [truncated]
```

[Tool: WebSearch — query: arXiv 2025 2026 token consumption coding agents SWE-bench input tokens per task energy]

Web search results for query: "arXiv 2025 2026 token consumption coding agents SWE-bench input tokens per task energy"

```
Links: [{"title": "Token Economics for LLM Agents: A Dual-View Study from Computing and Economics", "url": "https://arxiv.org/html/2605.09104v1"}, {"title": "The Best Programming Language for Tokenmaxxing: An Investigation of Coding Agent Behavior Across Programming Languages", "url": "... [truncated]
```

[Tool: Bash — Check how EcoLogits LLM method treats input tokens]

```
[output token] ... have relevant data and avoid outliers, we have filtered the dataset to keep only (1) the benchmark on NVIDIA H100 80GB HBM3 GPUs, (2) batch sizes no larger than 512 and (3) models no larger than 200B parameters. We approximate energy consumption per output token as a function of the number of activate parameters, denoted as  $\backslash(P_{\text{active}}\backslash)$  , and the batch size, denoted as ... [truncated]
```

[Tool: Bash — Read EcoLogits assumptions and Couch blog post]

Assumptions and limitations To be able to estimate environmental impacts of LLMs at inference we took the approach of modeling the key components that compose the service. In this section we will list major assumptions we make when modeling environmental impacts as well as known limitations. When possible we will try to quantify the potential inaccuracies. On models Two major information we are lo ... [truncated]

[Tool: Bash — Read remainder of Couch estimate and figures]

The assumption I'm going to make here is that energy usage for different kinds of tokens can be reasonably estimated by scaling according to the billing rates of those tokens. 2 We have $(input_tokens \times input_rate) + (output_tokens \times output_rate) = total_Wh$ and $output_rate = 5 \times input_rate$, and can substitute the second into the first to get: $input_rate = 2.5 / (0.01 + 5 \times 0.00053)$ $input_rate = 2. ... [truncated]$

[Tool: Bash — Read abstract and metadata of SWE-bench token paper]

```
How Do AI Agents Spend Your Money? Analyzing and Predicting Token Consumption in Agentic Coding Tasks  
['Bai, Longju', 'Huang, Zhemin', 'Wang, Xingyao', 'Sun, Jiao', 'Mihalcea, Rada',  
'Brynjolfsson, Erik', 'Pentland, Alex', 'Pei, Jiaxin']  
['2026/04/24']
```

```
Abstract:The wide adoption of AI agents in complex human workflows is driving rapid growth in LLM token consumption. When agents are d ... [truncated]
```

[Tool: Bash — Extract per-task token figures from SWE-bench paper]

70748
[M tokens] ... erty of the problem. Additionally, models generally consume more tokens on the shared failure subset than on the shared success subset, but the size of this gap varies substantially across models: GPT-5 and GPT-5.2 show only a mild increase (< < 0.5M tokens), whereas Kimi-K2 exhibits a much larger rise (~ \sim 2M tokens). One likely explanation is that models lack a reliable ... [truncated]

[Tool: Bash — List agent-session energy per model in calculator data]

GPT-5.5 chat 2.6601 agent 644.1723 [414.5847, 873.76]
GPT-5.5 Pro chat 101.6082 agent 23130.7433 [12110.535, 34150.952]
GPT-5.4 mini chat 0.3093 agent 76.688 [19.524, 133.8523]
Claude Opus 4.8 chat 1.8735 agent 459.042 [313.7117, 604.3727]
Claude Sonnet 4.6 chat 0.7811 agent 190.8147 [141.8623, 239.7673]
Claude Haiku 4.5 chat 0.0446 agent 10.9567 [6.038, 15.8757]
Gemini 3.1 Pro chat 5.3893 agent 1 ... [truncated]

Coding-agent notes are drafted below. The research turned up a **structural problem with the calculator's current agent figure**, which I've put first because it affects how you might approach Jordan's needs.

Draft: Coding agents and long, irregular sessions (serves Jordan)

Key finding: the current "agent session" is measured differently from how real sessions work

- **The calculator's current method.** It treats one agent session as **100,000 output tokens** and applies EcoLogits' per-output-token energy. That gives, for example, **~459 Wh** (range 314–604) for Claude Opus 4.8 and **~644 Wh** for GPT-5.5 ([data.js](#)).
- **EcoLogits counts only output tokens.** Its LLM method estimates energy as *number of output tokens × energy per output token* and does not model input tokens ([EcoLogits LLM method](#)).
- **Real agent sessions are dominated by input tokens.** Across 8 frontier models on 500 SWE-bench Verified coding tasks, agentic tasks used **~1,000× more tokens** than code chat, and **input tokens, not output, drove the cost**, even with caching (Bai et al. 2026, below).
- So the current figure could be too high or too low for a given person, because it counts the wrong kind of token. I can't say which direction without better per-token evidence.

Evidence (checked directly)

Source	What it shows
Bai, Huang, Wang, Sun, Mihalcea, Brynjolfsson, Pentland & Pei, "How Do AI Agents Spend Your Money?", arXiv, Apr 2026 (arXiv)	~1,000× more tokens than chat, driven by input tokens. Runs of the same task varied up to 30× in total tokens. Some models used over 1.5 million more tokens than others on the same tasks. More tokens did not mean more accuracy. Models underestimated their own token needs.
Couch, "Electricity use of AI coding agents," personal blog, 20 Jan 2026 (post)	From his own Claude Code logs (8,825 API calls): the median session had 24 model calls and ~592,000 tokens , mostly cache reads. He estimated ~41 Wh per median session and ~1,300 Wh on a median working day running 2–3 agents at once, which he likens to an extra dishwasher run per day.

For scale: Couch's 41 Wh session $\approx$ 15 GPT-5.5 chatbot replies at the calculator's figure, while his 1,300 Wh day $\approx$ ~500.

Limitations and contrary points

- **Couch calls his own figures "napkin math from Some Guy."** They are one person's usage on one model (Opus 4.5). His energy per token is derived from Epoch AI's GPT-4o estimate and scaled by *API prices*, assuming energy tracks price. That's a large, unverified assumption, and he says "I have no idea if these are reasonable guesses" about the cached-token figures. He deliberately used pessimistic long-context rates. *Proposed decision: use only as an illustration of session structure, not as a figure.*
- **Bai et al. is a preprint** that measures *tokens and dollars, not energy*. SWE-bench tasks are benchmark bug fixes, which may not resemble Jordan's creative-technology sessions.
- **No source I found measures energy per cached input token.** Caching reuses earlier computation, so it is presumably cheaper, but how much cheaper is not published. This is the biggest gap.
- **None of the providers publish per-session energy for their coding agents.**

What this means for Jordan's stated needs

- **"Prompts per day is too simple":** supported. One user message sets off many model calls (Couch's median session was 24 calls for 5 user messages), and the same task can vary 30x.
- **Wants assumptions, ranges and project totals:** the evidence supports wide ranges, and an input for session count or length rather than one fixed session size. Any per-session figure would have to be shown as highly uncertain.

Assumptions still to check

- Energy per input token and per cached token. I'd look next at ML.Energy and the Jegham et al. 2025 "How Hungry is AI?" study, which I haven't checked yet.
- How long and how frequent Jordan-like sessions are in practice (no evidence yet beyond one blogger).
- Whether Anthropic's own documentation gives average daily usage for its coding tool that could serve as a second data point. Not checked yet.

Question: Should I add this to [research.md](#) now and come back to the per-input-token energy gap later, or chase that gap first? It's the piece that decides whether a better agent estimate is even possible.

8. Learner (RG Sureshkumar)

chase the per input token energy gap first

Agent

[Tool: WebSearch — query: energy per token prefill versus decode measured input tokens output tokens LLM inference joules arXiv]

Web search results for query: "energy per token prefill versus decode measured input tokens output tokens LLM inference joules arXiv"


```
Links: [{"title": "Characterization of Request and Token Energy Costs for LLM Inference Workloads on GPU Platforms", "url": "https://arxiv.org/pdf/2608.28044"}, {"title": "Characterization of Request and Token Energy Costs for LLM Inference Workloads on GPU Platforms", " ... [truncated]
```

[Tool: WebSearch — query: prompt caching KV cache reuse energy savings measurement LLM inference prefix caching joules]

Web search results for query: "prompt caching KV cache reuse energy savings measurement LLM inference prefix caching joules"

```
Links: [{"title": "Prompt Caching Infrastructure | IntroL Blog", "url": "https://introl.com/blog/prompt-caching-infrastructure-llm-cost-latency-reduction-guide-2025"}, {"title": "KV cache: How to speed up LLM inference (part 1)", "url": "https://datasciencedojo.com/blog/kv-cache-h ... [truncated]
```

[Tool: Bash — Extract Epoch AI input-length energy estimates]

```
36802
[Published] ... pics Leading companies Energy Cite Epoch AI's work is free to use, distribute, and reproduce provided the source and authors are credited under the Creative Commons Attribution license . Citation Josh You (2025), "How much energy does ChatGPT use?". Published online at epoch.ai. Retrieved from 'https://epoch.ai/gradient-updates/how-much-energy-does-chatgpt-use' [online resour ... [truncated]
```

[Tool: Bash — Fetch abstracts of five per-token energy papers]

```
# 2608.28044 Characterization of Request and Token Energy Costs for LLM Inference Workloads on GPU Platforms
['Vellaisamy, Prabhu', 'Lam, Vanessa', 'Blanton, Shawn', 'Shen, John Paul'] ['2026/08/28']
Abstract:Large language model (LLM) inference serving is priced by tokens, but GPU energy is consumed over inference windows. This accounting mismatch makes token-normalized metrics incomplete, since ... [truncated]
```

[Tool: Bash — Extract input-length and caching energy figures from three papers]

```
##### 2505.09598
71717
[10k] ... Energy Consumption Figure 3 : Energy consumption per model across three prompt sizes (Wh, log-scale). Model Energy Consumption (100 input-300 output) (Wh) Energy Consumption (1k input-1k output) (Wh) Energy Consumption (10k input-1.5k output) (Wh) GPT-4.1 0.871 ± 0.302 3.161 ± 0.515 4.833 ± 0.650 GPT-4.1 mini 0.450 ± 0.081 1.545 ± 0.211 2.122 ± 0.348 GPT-4.1 nano ... [truncated]
```

[Tool: Bash — Search ML.Energy paper for prefill and input-length findings]

```
52745
[prefill] ... n. This made FP8 consistently consume more time and energy across all batch sizes. Attention data parallelism incurs load imbalance between GPUs that are assigned very different sequence lengths (e.g., some running long prefills whereas others run decode). Since
```

the straggler GPU bottlenecks the entire batch, this can lead to significant latency overhead. Furthermore, the non-s ... [truncated]

[Tool: WebSearch — query: measured energy consumption agentic coding session SWE-bench Wh per task GPU measurement 2026]

Web search results for query: "measured energy consumption agentic coding session SWE-bench Wh per task GPU measurement 2026"

Links: [{"title": "AgentStop: Terminating Local AI Agents Early to Save Energy in Consumer Devices", "url": "https://arxiv.org/pdf/2605.15206"}, {"title": "ProgRouter: Online Progress-Guided Orchestration for Multi-Agent LLM Workflows under Quality-Cost Tradeoffs", "url": "https: ... [truncated]

Quick update: I've finished checking the evidence on per-token energy. The gap can be narrowed but not closed. Findings below.

Per-input-token energy: what the evidence can and can't support

Bottom line

- **No public source measures energy per input token, or per cached input token, for the frontier cloud models people actually use for coding.**
- Three *estimation* methods exist, and for **fresh input tokens** they roughly agree to within a few times.
- For **cached input tokens**, which make up most of an agent session, **there is no evidence-based figure at all**. Only a price-based guess exists.

What each method implies for ~10,000 extra input tokens

Source	Method	Implied energy for 10k input tokens
Josh You, "How much energy does ChatGPT use?", Epoch AI, 2025 (article)	Compute model for GPT-4o from assumed parameter count, hardware and utilisation. Self-described as "somewhat pessimistic."	~2.1 Wh : a 10k-input query is ~2.4 Wh vs ~0.3 Wh for a typical one. A 100k-input query is ~40 Wh, because cost grows roughly with the square of input length.
Jegham, Abdelatti, Koh, Elmoubarki & Hendawi, "How Hungry is AI?", arXiv 2025, v6 (arXiv)	Inferred from public API speed data plus assumed hardware; not directly metered	Under ~1.7 Wh . GPT-4.1 goes from 3.16 Wh (1k input, 1k output) to 4.83 Wh (10k input, 1.5k output), and that increase also includes 500 more output tokens.
Couch 2026 (already noted)	API prices used as a stand-in for energy	~2 to 4 Wh , depending on which context-length rate is used

Cached tokens: - Couch assumes cached reads cost 1/10 the energy of fresh input, only because Anthropic charges 1/10. He says he has "no idea" whether that's reasonable. - A search result claimed "cache hits use 90% less energy." It traced to a vendor blog with no measurement behind it. **I propose rejecting it.** - The only peer-reviewed-style evidence is qualitative. Caching avoids recomputation and saves operating energy, but the storage hardware adds manufacturing carbon ([Tian et al. 2025, "GreenCache," arXiv](#)).

Why the measured lab studies don't fill the gap

- **The strongest measurement studies use small open models.** ML.Energy (Chung et al. 2026, [arXiv](#)) and Vellaisamy et al. (IISWC 2026, [arXiv](#)) both do.
- **Vellaisamy et al. show that "energy per token" is itself a slippery unit.** Longer outputs lowered joules per token while *raising* total energy (7.46 → 0.72 J/token, but 1.19 → 5.93 kJ total). One fixed per-token rate hides that.
- **Studies that measure energy for whole agent tasks run small local models on consumer or workstation GPUs** (for example SWEnergy, [arXiv](#); AgentStop, [arXiv](#)). I read only the titles and abstracts, and they don't represent cloud frontier models.

What this means for the calculator (options for you, not decisions)

1. **Keep the current output-only EcoLogits figure** and clearly disclose that it ignores input tokens and doesn't reflect how real sessions are structured.
2. **Add an input-token component** using the ~1–4 Wh per 10k fresh tokens range, and show cached tokens as an explicitly unknown range rather than a single figure.
3. **Show a session-level range** built from the structure evidence (Bai et al. on variability, Couch as a worked example), clearly labelled as an illustration.

Any of these would meet Jordan's stated need for visible assumptions and ranges. Options 2 and 3 introduce more uncertainty that has to be communicated.

Proposed source decisions: - **Use with qualifications:** Epoch AI and Jegham 2025. - **Background only, for mechanism:** Chung, Vellaisamy and GreenCache. - **Reject:** the "90% less energy" cache claim.

Question: Should I now add the whole coding-agents section to [research.md](#), including this per-token analysis and the proposed source decisions (as S7 onward)?

9. Learner (RG Sureshkumar)

yes add it to research.md

Agent

[Tool: Bash — Check the vendor blog's cache energy claim]

60

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

I've added the coding-agents section to [research.md](#) as a draft.

- **Audience needs:** a new "3. Coding agents and long, irregular sessions (Jordan)" section. It covers:
 - the mismatch between the calculator's output-only figure and real sessions, which are dominated by input tokens
 - the ~1–4 Wh per 10,000 fresh input tokens range
 - the lack of any evidence-based figure for cached tokens
 - the three calculator options, recorded explicitly as yours to decide
- **Source assessments:** S7 to S13.
 - S7: Bai et al.
 - S8: Couch, proposed as *illustration only*
 - S9: EcoLogits LLM method
 - S10: Epoch AI
 - S11: Jegham 2025
 - S12: background mechanism sources
 - S13: the rejected "90% less energy" cache claim

One thing to know about S13: the vendor page wouldn't load in readable form, so the rejection rests on finding no primary measurement behind the claim. I didn't read the page itself, and the entry says so.

Everything is still a proposal for you to confirm. Nothing is marked approved.

Next on the list is **4. Occasional text AI (Robin)**. That's already in the calculator, so this would be a check of whether the EcoLogits per-prompt figures agree with the disclosures and estimates published since. Examples are Google's figure for a median Gemini prompt and Mistral's life-cycle report, both of which include water.

Question: Should I go ahead with occasional text AI?

10. Learner (RG Sureshkumar)

yes lets move on to occasional text AI

Agent

[Tool: Bash — Read Google Gemini environmental impact paper]

```
Measuring the environmental impact of delivering AI at Google Scale
['Elsworth, Cooper', 'Huang, Keguo', 'Patterson, David', 'Schneider, Ian', 'Sedivy, Robert',
'Goodman, Savannah', 'Townsend, Ben', 'Ranganathan, Parthasarathy', 'Dean, Jeff', 'Vahdat,
Amin', 'Gomes, Ben', 'Manyika, James'] ['2025/08/21']
```

Abstract:The transformative power of AI is undeniable – but as user adoption accelerates, so ... [truncated]

[Tool: WebSearch — query: Mistral AI life cycle analysis Mistral Large 2 400 token response 1.14 g CO₂e 45 mL water]

Web search results for query: "Mistral AI life cycle analysis Mistral Large 2 400 token response 1.14 g CO₂e 45 mL water"

Links: [{"title":"DeepLearning.AI on X: \"Mistral published an 18-month life-cycle analysis of Mistral Large 2. The study measures greenhouse-gas emissions, energy use, and consumption of water and other materials across data-center construction, hardware manufacturing, traini ... [truncated]"}]

[Tool: WebSearch — query: critique Google Gemini 0.24 Wh median prompt water 0.26 mL market-based emissions Shaolei Ren]

Web search results for query: "critique Google Gemini 0.24 Wh median prompt water 0.26 mL market-based emissions Shaolei Ren"

Links: [{"title":"Google: Median Gemini prompt uses 0.24 watt hours of power and consumes 0.26ml of water – DCD","url":"https://www.datacenterdynamics.com/en/news/google-median-gemini-prompt-uses-024-watt-hours-of-power-and-consumes-026ml-of-water/"},"{"title":"Experts accu ... [truncated]"}]

[Tool: Bash — Verify Mistral, Altman, and critique sources directly]

MISTRAL

LEN 10652

[1.14] ... impacts: 20,4 ktCO₂e, 281 000 m³ of water consumed, and 660 kg Sb eq (standard unit for resource depletion). the marginal impacts of inference, more precisely the use of our AI assistant Le Chat for a 400-token response – excluding users' terminals: 1.14 gCO₂e, 45 mL of water, and 0.16 mg of Sb eq. These figures reflect the scale of computation involved in Gen AI, ... [truncated]

[Tool: WebSearch — query: The Verge Google Gemini five drops of water experts misleading Shaolei Ren Alex de Vries]

Web search results for query: "The Verge Google Gemini five drops of water experts misleading Shaolei Ren Alex de Vries"

Links: [{"title":"Experts accuse Google of downplaying AI water use with '5 drops per prompt' claim – Neowin","url":"https://www.neowin.net/news/experts-accuse-google-of-downplaying-ai-water-use-with-5-drops-per-prompt-claim/"},"{"title":"Experts are skeptical about Google's AI ... [truncated]"}]

[Tool: Bash — Read PCWorld coverage of Google water critique]

LEN 5321

[Verge] ... tock.com Yesterday, we covered Google's report that a typical query to its Gemini

AI consumes only “five drops of water.” That figure is now facing criticism from several AI experts, according to The Verge ... and that includes one of the authors of one of the reports referred to by Google. AI researcher Shaolei Ren –a professor at University of California Riverside and one of t ... [truncated]

[Tool: Bash — List per-prompt energy and water for short text in calculator]

GPT-5.5	tweet	0.406Wh	1.47mL		chat	2.66Wh	(1.74–3.58)	9.6mL
GPT-5.5 Pro	tweet	20.758Wh	74.86mL		chat	101.61Wh	(57.53–145.69)	366.4mL
GPT-5.4 mini	tweet	0.041Wh	0.15mL		chat	0.31Wh	(0.08–0.54)	1.1mL
Claude Opus 4.8	tweet	0.267Wh	1.00mL		chat	1.87Wh	(1.29–2.46)	7.0mL
Claude Sonnet 4.6	tweet	0.114Wh	0.42mL		chat	0.78Wh	(0.58–0.98)	2.9mL
Claude Haik ...	[truncated]							

I've finished checking the text-AI sources: Google's paper, Mistral's life-cycle report, OpenAI's figure, and the published criticism. I've also compared them with the calculator's current numbers. Draft below.

Draft: Occasional text AI (serves Robin)

Main finding

- Published figures for a "typical prompt" cluster around 0.24–0.42 Wh.
- The calculator's EcoLogits figures for a 300-word reply from flagship models are several times higher, for example ~2.7 Wh for GPT-5.5 and ~5.4 Wh for Gemini 3.1 Pro.
- Water figures differ by more than 100x, depending almost entirely on what each source chooses to count.
- This doesn't show the calculator is wrong. The numbers measure different things. But Robin would reasonably ask why they don't match, so the calculator needs to be able to explain it.

Evidence (checked directly)

Source	Figure	What it counts
Elsworth et al. (Google, incl. Patterson and Dean), "Measuring the environmental impact of delivering AI at Google Scale," arXiv, Aug 2025 (arXiv)	Median Gemini Apps text prompt (May 2025): 0.24 Wh , 0.03 g CO₂e , 0.26 mL water . Energy per prompt fell 33x over one year.	<i>Measured</i> in production: chips, host machines, idle capacity and data-center overhead. Median across all Gemini models. Carbon is market-based , meaning it's reduced by Google's clean-energy purchases. Water is on-site cooling only . Excludes training, networking and user devices.
Altman, "The Gentle Singularity," personal blog, June 2025 (post)	"Average" ChatGPT query ~0.34 Wh , ~0.000085 gallons (~0.32 mL) of water	No method given.
Mistral AI with Carbone 4 and ADEME, life-cycle report, 22 July 2025 (post)	Le Chat, 400-token response: 1.14 g CO₂e , 45 mL water	Full life cycle , including manufacturing ("upstream") impacts. Reviewed by two environmental consultancies. No Wh figure.
Epoch AI (S10) and Jegham 2025 (S11)	GPT-4o typical query ~0.3 Wh (Epoch); 0.42 Wh short prompt (Jegham)	Modelled or inferred, not measured

Source	Figure	What it counts
Calculator (EcoLogits, S9)	300-word reply: 0.04 Wh (Haiku 4.5) to 5.4 Wh (Gemini 3.1 Pro); GPT-5.5 2.7 Wh, 9.6 mL	Modelled per model. Water includes on-site plus power-plant water (Li et al. method).

Criticism and contrary evidence

- **Experts disputed Google's figures on two points**, as reported by The Verge. I verified this through [PCWorld's coverage](#), 22 Aug 2025, because the Verge article itself didn't load.
- It **leaves out water used by power plants**.
- It **uses only market-based carbon** rather than the local grid mix.
- **Shaolei Ren (UC Riverside)**, whose earlier estimates Google compared itself against, said Google was "hiding the critical information."
- **Google's paper hadn't been peer reviewed** when published.
- **In fairness to Google**, the paper argues a *median* is more representative than a mean, because a small share of heavy prompts skews the average.

Why the numbers disagree (for Robin's explanation)

1. **Which prompt:** Google's median mixes many models, including small ones. The calculator shows specific flagship models at a fixed reply length.
2. **What's measured:** Google's figure is measured in production. EcoLogits, Epoch and Jegham model or infer.
3. **What's counted for water:** on-site cooling only (0.26 mL), on-site plus power plants (EcoLogits ~1–22 mL for a 300-word reply), or the full life cycle including manufacturing (Mistral, 45 mL).
4. **How carbon is counted:** market-based (lowered by clean-energy purchases) vs. location-based (the actual grid).
5. **When:** figures change fast. Google reports a 33x drop in one year, so 2025 figures may not describe 2026 models.

Assumptions still to check

- **How the reply length and model Robin would realistically use relate to a "median prompt."**
- **Whether EcoLogits tends to estimate high for today's models.** I found no direct validation against measured production data.
- **Andy Masley**, who wrote the original calculator, published a rebuttal to some of the criticism of Google. I haven't read it. Because the calculator is adapted from his work, it would need careful handling if we use it.

Proposed source decisions: - **Google:** use with qualifications, stating its boundaries explicitly. - **Mistral:** use with qualifications, as the full life-cycle comparison point. - **Altman:** use only as a stated company claim with no method, or reject. Your call. - **PCWorld/Verge criticism:** use as documented disagreement.

Question: Should I add this to [research.md](#) as section 4 with sources S14 onward?

11. Learner (RG Sureshkumar)

yes add it to research.md

Agent

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

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I've added the occasional text AI notes to [research.md](#) as a draft.

- **Audience needs:** a new "4. Occasional text AI (Robin)" section. It covers:
 - the published "typical prompt" figures against the calculator's figures
 - the 100x-plus spread in water depending on what is counted
 - a five-point explanation of why the figures disagree, which could feed a plain-language explanation for Robin
 - the published criticism
- **Source assessments:** S14 Google, S15 Altman, S16 Mistral and S17 the criticism coverage.

Two notes on this section: - S17 relies on PCWorld's report of The Verge's article, because the original wouldn't load. The quotation from Shaolei Ren is therefore secondhand, and the entry says so. - For S15 (Altman), I left "use as an unsupported company claim, or reject" as your decision.

That completes the professional AI activities (1 to 4). Next is the first "broader digital life" activity, **5. Video calls (Jordan, Robin)**. I'd look for energy and data estimates per hour of video calling, including how much depends on the user's device and network versus the data center, and whether camera-off or lower resolution changes much.

Question: Should I go ahead with video calls?

12. Learner (RG Sureshkumar)

yes move on to video calls

Agent

[Tool: WebSearch — query: Obringer 2021 overlooked environmental footprint increasing internet use videoconference camera off 96% Resources Conservation Recycling]

Web search results for query: "Obringer 2021 overlooked environmental footprint increasing internet use videoconference camera off 96% Resources Conservation Recycling"

Links: [{"title": "The overlooked environmental footprint of increasing Internet use | Request PDF", "url": "https://www.researchgate.net/publication/348414802_The_overlooked_environmental_footprint_of_increasing_Internet_use"}, {"title": "... [truncated]

[Tool: WebSearch — query: video conferencing energy consumption per hour Wh measurement study devices network data centre Zoom Teams lifecycle]

Web search results for query: "video conferencing energy consumption per hour Wh measurement study devices network data centre Zoom Teams lifecycle"

Links: [{"title": "Can You See Me Now? A Measurement Study of Zoom, Webex, and Meet", "url": "https://arxiv.org/pdf/2109.13113"}, {"title": "Hourly Data Consumption of Popular Video Conferencing Applications - CableLabs", "url": "https://www.cablelabs.com/b ... [truncated]

[Tool: WebSearch — query: IEA Kamiya energy intensity data transmission kWh per GB critique Obringer overestimate]

Web search results for query: "IEA Kamiya energy intensity data transmission kWh per GB critique Obringer overestimate"

Links: [{"title": "Network energy use not directly proportional to data volume", "url": "https://ora.ox.ac.uk/objects/uuid:1ac080d9-5a51-49a8-b153-b70ec565ff4f/files/rbk128b95s"}, {"title": "The carbon footprint of streaming video: fact-checking the headlines - Analysis - IEA", "url": "... [truncated]

[Tool: Bash — Read Obringer press release and Mytton critique]

SCIENCEDAILY
LEN 13768

[96] ... dioxide (a gallon of gasoline burned from a car emits about 8,887 grams), requires 2–12 liters of water and demands a land area adding up to about the size of an iPad Mini. But leaving your camera off during a web call can reduce these footprints by 96%. Streaming content in standard definition rather than in high definition while using apps such as Netflix or H ... [truncated]

[Tool: Bash — Read Mytton 2024 abstract and IEA streaming commentary]

WILEY MYTTON
LEN 60
CROSSREF

['Network energy use not directly proportional to data volume: The power model approach for more reliable network energy consumption calculations'] ['David Mytton', 'Dag Lundén', 'Jens Malmodin'] ['Journal of Industrial Ecology'] {'date-parts': [[2024, 6, 21]]} 28 4 966–980

Abstract

It is commonly assumed that data volume and network energy co ... [truncated]

[Tool: WebFetch — <https://www.iea.org/commentaries/the-carbon-footprint-of-streaming-video-fact-checking-the-headlines>]

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

[Tool: Bash — Read CableLabs data use and Greenspector device energy]

```
### CABLELABS
LEN 12116
[GB] ... ar to the left is the lower-cost laptop and the bar to the right is the higher-cost
laptop. Key Observations A. Data Consumption Varies : The first takeaway is that different
apps consume different amounts of bandwidth, as shown in Table 1, from 0.5 GBytes per hour up
to 3.4 GBytes per hour, for video conferences using the different laptops, the same broadband
conn ... [truncated]
```

[Tool: Browser navigate — <https://www.iea.org/commentaries/the-carbon-footprint-of-streaming-video-fact-checking-the-headlines>]

```
{
  "serverId": "preview-local_d8c13413-de50-49cf-a3bc-5e5facf737ed",
  "tabId": "seed",
  "reused": false,
  "type": "browser",
  "navOk": true
}
Browser pane opened. Use serverId "preview-local_d8c13413-de50-49cf-a3bc-5e5facf737ed" with
read_page / computer / navigate.
```

Quick update: the IEA blocks automated downloads, so I've opened its streaming-energy commentary in the browser pane to read it directly.

[Tool: Browser get_page_text —]

```
Title: Just a moment...
URL: https://www.iea.org/commentaries/the-carbon-footprint-of-streaming-video-fact-checking-
the-headlines
Source element: <div>
---
www.iea.org
Performing security verification

This website uses a security service to protect against malicious bots. This page is displayed
while the website verifies you are not a bot.

Tab Context:
```

```
- Executed on tabId: seed
- Available tabs ... [truncated]
```

[Tool: Browser computer — wait]

```
waited 5s
```

```
Tab Context:
```

```
- Executed on tabId: seed
- Available tabs:
  • tabId seed: "Just a moment..." ("https://www.iea.org/commentaries/the-carbon-footprint-of-streaming-video-fact-checking-the-headlines")
```

[Tool: Browser get_page_text —]

```
Title: Just a moment...
```

```
URL: https://www.iea.org/commentaries/the-carbon-footprint-of-streaming-video-fact-checking-the-headlines
```

```
Source element: <div>
```

```
---
```

```
www.iea.org
```

```
Performing security verification
```

```
This website uses a security service to protect against malicious bots. This page is displayed while the website verifies you are not a bot.
```

```
Tab Context:
```

```
- Executed on tabId: seed
- Available tabs ... [truncated]
```

[Tool: Bash — Read archived copy of IEA streaming commentary]

```
LEN 25446
```

```
[kWh] ... that watching a YouTube video uses over 1600W of electricity, equivalent to the consumption of 15 big screen TVs Looking at electricity consumption alone, the original Shift Project figures imply that one hour of Netflix consumes 6.1 kilowatt hours (kWh) of electricity. This is enough to drive a Tesla Model S more than 30km , power an LED lightbulb constantly for a month, or bo ... [truncated]
```

[Tool: Bash — Extract IEA device and network energy breakdown]

```
r boil a kettle once a day for nearly three months. The corrected figures imply that one hour of Netflix consumes 0.8 kWh. With 167 million Netflix subscribers watching an average of 0.8 kWh of electricity. This is enough to drive a Tesla Model S more than 30km , power an LED lightbulb constantly for a month, or boil a kettle once a day for nearly three months. The corrected figures imply t ... [truncated]
```

Video call notes are drafted below. **The evidence here is weaker than for the AI activities.** I found no recent, peer-reviewed figure for energy per hour of video calling. The most-quoted figure (from Obringer et al.) rests on assumptions that later research rejects.

The IEA's site blocked access with a bot check, which I didn't try to get past. I read an Internet Archive copy of the page instead.

Draft: Video calls (serves Jordan and Robin)

Evidence (checked directly)

Source	Figure	Notes
Obringer, Rachunok, Maia-Silva, Arbabzadeh, Nateghi & Madani, "The overlooked environmental footprint of increasing Internet use," <i>Resources, Conservation and Recycling</i> 167:105389, 2021 (DOI); verified through Purdue's press release)	One hour of videoconferencing or streaming: 150–1,000 g CO₂ . Camera off cuts it by 96% .	Widely quoted. Covers data transmission and data centers per gigabyte, not the user's own device .
Mytton, Lundén & Malmodin, "Network energy use not directly proportional to data volume," <i>Journal of Industrial Ecology</i> 28(4):966–980, 2024 (DOI)	Peer-reviewed argument that " kWh per gigabyte " methods are insufficient for real network energy. Network energy barely changes when one person uses more or less data.	This is the central contrary evidence. It undermines per-GB methods like Obringer's, which is where the 96% figure comes from.
Mytton, blog notes on Obringer et al. (post)	Obringer used 0.06 kWh/GB for networks (a 2015 value) versus the IEA's 0.002 kWh/GB , and a data-center figure whose source he couldn't trace	A blog post by a co-author of the paper above, so not independent
Kamiya, IEA commentary, 10 Dec 2020 (page , read via archive copy)	<i>Streaming</i> , not calls: ~0.077 kWh per hour in 2019. A 50-inch TV uses ~100× a phone's electricity and ~5× a laptop's. Making the device accounts for ~80% of a phone's lifetime carbon.	Relevant for how devices dominate. Also key for activity 6 (streaming).
Jones & Zhu, CableLabs, 6 May 2021 (post)	Video calls used 0.5 to 3.4 GB per hour depending on the app (Meet, GoToMeeting, Teams, Zoom)	Industry lab test. Data volume only.
Greenspector, 6 Sept 2022 (post)	10 apps measured on a phone and a PC: camera on ~2.1× the phone battery energy of audio only. 61% of carbon from the user's device . Average ~0.66 g CO ₂ e per minute.	Commercial consultancy. One-to-one calls, 1-minute tests. The method isn't fully published.

What the evidence suggests

- **The user's own device probably matters most.** Greenspector puts it at 61% of the carbon, and the IEA shows device choice driving streaming energy. Laptop vs. desktop vs. large meeting-room screen would matter more than the app.
- **Turning the camera off saves something, but the "96%" figure is likely overstated.** It depends on the per-GB network method that later research rejects. Greenspector's device measurements suggest roughly half the device energy on a phone, with the method caveats above.
- **Data volume varies about 7× by app** (CableLabs). Under the newer power-model view, that barely changes network energy.

Limitations

- **No peer-reviewed, recent, whole-system energy figure per hour of video calling was found.**
- **Group calls aren't covered:** most tests are one-to-one, but every participant's device and connection adds energy.
- **Sources date from 2020–22.** Devices, codecs and networks have become more efficient since.

Assumptions still to check

- **Power draw of typical laptops, desktops and meeting-room displays** during a call. That overlaps with activity 9 (devices), and is needed to build any estimate for Jordan and Robin.
- **Whether video-call providers such as Zoom, Microsoft or Google publish per-hour figures.** Not checked yet.
- **Whether a calculator should count the device at all**, given people would own it anyway. That's a methodology choice for you.

Proposed source decisions: - **Obringer:** use only with strong qualifications, or reject for the 96% claim. - **Mytton et al. 2024:** use. - **IEA:** use with qualifications for device comparisons. - **CableLabs:** use for data volume only. - **Greenspector:** use with qualifications. - **Mytton's blog notes:** background only.

Question: Should I add this to [research.md](#) as section 5, or first check whether Zoom, Microsoft or Google publish their own per-hour figures?

13. Learner (RG Sureshkumar)

check on the per hour figures for each of these

Agent

[Tool: WebSearch — query: Zoom carbon footprint per meeting hour sustainability report energy per meeting minute]

Web search results for query: "Zoom carbon footprint per meeting hour sustainability report energy per meeting minute"

Links: [{"title": "Green Tips: Zoom consumes energy; here's how to cut back – Park Record", "url": "https://www.parkrecord.com/2025/10/22/green-tips-zoom-consumes-energy-heres-how-to-cut-back/"}, {"title": "Just How Green is Video Conferencing? New Study Explores the Environmental Imp ... [truncated]"}]

[Tool: WebSearch — query: Microsoft Teams meeting carbon emissions per hour Microsoft study energy videoconference]

Web search results for query: "Microsoft Teams meeting carbon emissions per hour Microsoft study energy videoconference"

Links: [{"title": "Assessing the Carbon Footprint of Virtual Meetings: A Quantitative Analysis"}]

of Camera Usage","url":"https://arxiv.org/html/2601.06045v2"}, {"title": "Assessing the Carbon Footprint of Virtual Meetings: A Quantitative Analysis of", "url": "https://arxiv.org/pdf/260 ... [truncated]

[Tool: WebSearch — query: Google Meet carbon footprint per hour video call Google sustainability disclosure]

Web search results for query: "Google Meet carbon footprint per hour video call Google sustainability disclosure"

Links: [{"title": "The hidden pollution cost of online meetings | Gerry McGovern", "url": "https://gerrymcgovern.com/the-hidden-pollution-cost-of-online-meetings/"}, {"title": "The Invisible Carbon Footprint: How Unnecessary Meetings Are Harming Our Planet | meetingcosts.com", "url": "https: ... [truncated]

[Tool: Bash — Check Zoom report, Microsoft Q&A, and 2026 camera study]

ZOOM
LEN 8659

MS QA
LEN 4156

[camera] ... What is the difference in Co2 emissions between a Teams meeting with camera's ON vs camera's OFF ? – Microsoft Q&A Skip to main content This browser is no longer supported. Upgrade to Microsoft Edge to take advantage of the latest features, security updates, and technical support. Download Microsoft Edge More info about Internet Explorer and Micr ... [truncated]

[Tool: WebFetch — https://www.zoom.com/en/about/esg-impact-report/]

Based on the provided webpage content, **no figures are given for energy use or carbon emissions per meeting, per minute, per hour, or per participant of a Zoom meeting.**

Regarding company-level emissions figures, the page does not include specific emissions data or a report year for such metrics. The environmental impact section mentions achievements like "Global renewable energy coverage of 94 ... [truncated]

[Tool: Bash — Read full Microsoft Q&A thread text]

Hoping someone in microsoft has done this testing, if not could you provide the metrics for CO2 used when camera's are on vs when cameras are off? Microsoft Teams | Microsoft Teams for business | Meetings and calls | Audio and video Microsoft Teams | Microsoft Teams for business | Meetings and calls | Audio and video Managing sound and video settings during Teams meetings and calls for optimal com ... [truncated]

[Tool: WebSearch — query: Zoom FY26 Impact Report PDF emissions per meeting minute]

Web search results for query: "Zoom FY26 Impact Report PDF emissions per meeting minute"

Links: [{"title": "Zoom Releases FY 2026 Impact Report | Zoom", "url": "https://news.zoom.com/zoom-releases-fy-2026-impact-report/"}, {"title": "© 2025 Zoom Communications, Inc. Zoom Communications Q3 FY26

```
Earnings", "url": "https://investors.zoom.us/static-files/86a5154d-f068-48e9-947c-064395a9d786"}, {"title": "Carb ... [truncated]}
```

[Tool: Bash — Find Zoom FY26 Impact Report PDF link]

```
href="https://news.zoom.com/wp-json/oembed/1.0/embed?url=https%3A%2F%2Fnews.zoom.com%2Fzoom-releases-fy-2026-impact-report%2F"
href="https://news.zoom.com/wp-json/oembed/1.0/embed?url=https%3A%2F%2Fnews.zoom.com%2Fzoom-releases-fy-2026-impact-report%2F&#038;format=xml"
href="https://news.zoom.com/zoom-releases-fy-2026-impact-report/"
href="https://twitter.com/intent/tweet?text=&url=https://new ... [truncated]
```

I've finished checking Zoom, Microsoft and Google. **None of the three publishes an energy or carbon figure per hour of video calling that I could find.**

Provider	What I checked	Result
Zoom	Its Impact / ESG page and the FY2026 Impact Report announcement	No per-meeting, per-minute or per-hour figure. The page gives only company-level claims, such as 94% renewable energy coverage with its colocation data-center partner Equinix. It points to a full FY26 report that I couldn't find as a downloadable document, so I can't rule out a figure in there.
Microsoft (Teams)	Web search, and a Microsoft Q&A thread where a customer asked exactly this	No figure. The thread has no answer with Microsoft data. The person who asked concluded Microsoft hadn't shared any.
Google (Meet)	Web search for a Google disclosure	No figure found. Google's AI paper (S14) covers Gemini text prompts only.

A caution about search results: they show "50 g CO₂ per hour of Zoom/Teams" and "0.513 g per minute for Teams," but these aren't provider figures. - The 50 g figure comes from IdleForest, a browser-extension company whose method I haven't seen. I'd propose not using it unless we can verify it. - The Teams per-minute figure is Greenspector's measurement (already in the draft), not Microsoft's.

One new source turned up that bears on the camera question: - **Mortas, "Assessing the Carbon Footprint of Virtual Meetings: A Quantitative Analysis of Camera Usage,"** [arXiv:2601.06045](#), Dec 2025 ([arXiv](#)).

Accepted as a short paper at IARIA GREEN 2025. - Measured on a phone over 4G, it found that turning the camera off **roughly halves data use and the associated emissions**. The author says this *challenges* widely repeated claims, which implicitly includes Obringer's 96%. - **Limitations:** 4 pages, a single author, one phone on one mobile network, and it measures data, not energy. That per-data method is exactly what Mytton et al. 2024 criticise. - **Proposed decision:** use with qualifications as contrary evidence on the camera claim, not as a figure.

What this means for section 5: - "No provider discloses per-hour figures" becomes a recorded limitation. - The evidence against the 96% claim now has two sources that point the same way: Greenspector (~2× device energy on a phone) and Mortas (~2× data).

Question: Should I add the video calls section to [research.md](#) now, including these provider checks and the Mortas paper, as sources S18 onward?

14. Learner (RG Sureshkumar)

yes add it to research.md

Agent

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

I've added the video calls notes to [research.md](#) as a draft.

- **Audience needs:** a new "5. Video calls (Jordan, Robin)" section. It covers:
 - the disputed 150–1,000 g and 96% figures
 - the evidence that the camera-off saving is closer to half than 96%
 - the user's device as the probable main factor
 - the fact that Zoom, Microsoft and Google publish no per-hour figure, with the IdleForest "50 g" figure flagged and not proposed for use
- **Source assessments:** S18 to S24.
 - S18: Obringer, proposed for rejection as a calculator figure
 - S19: Mytton et al. 2024
 - S20: Mytton's blog notes
 - S21: IEA
 - S22: CableLabs
 - S23: Greenspector
 - S24: Mortas

Where my checks relied on indirect copies: - **S18:** confirmed through Purdue's press release, because the paper was paywalled. - **S19:** confirmed through its abstract only. - **S21:** read from an Internet Archive copy.

Everything is still a proposal for you to confirm. Nothing is marked approved. The question of whether to count the user's device is recorded as your decision.

Next is **6. Video and music streaming (Alex, Robin)**, on phone, laptop and TV. The IEA source (S21) already covers much of this. I'd add newer and independent estimates, such as the Carbon Trust's 2021 study, Netflix's own disclosures, and anything on music streaming, and check them against the network-energy debate.

Question: Should I go ahead with streaming?

15. Learner (RG Sureshkumar)

yes go ahead with streaming

Agent

[Tool: WebSearch — query: Carbon Trust 2021 carbon impact of video streaming white paper 55 gCO2e per hour Europe DIMPACT]

Web search results for query: "Carbon Trust 2021 carbon impact of video streaming white paper 55 gCO2e per hour Europe DIMPACT"

Links: [{"title": "Carbon Trust releases white paper on carbon impact of video streaming – Telecompaper", "url": "https://www.telecompaper.com/news/carbon-trust-releases-white-paper-on-carbon-impact-of-video-streaming--1458522"}, {"title": "Carbon Trust: 1hr streaming = 3 boi ... [truncated]"}]

[Tool: WebSearch — query: Netflix environmental social governance report carbon footprint one hour of streaming grams CO2e DIMPACT]

Web search results for query: "Netflix environmental social governance report carbon footprint one hour of streaming grams CO2e DIMPACT"

Links: [{"title": "The Carbon Cost of Streaming – Greenly", "url": "https://greenly.earth/en-us/leaf-media/data-stories/the-carbon-cost-of-streaming"}, {"title": "Is Netflix Succeeding in Making Streaming Greener? | Sustainability Magazine", "url": "https://sustainabil ... [truncated]"}]

[Tool: WebSearch — query: music streaming energy carbon footprint per hour study Spotify peer-reviewed]

Web search results for query: "music streaming energy carbon footprint per hour study Spotify peer-reviewed"

Links: [{"title": "Spotify's rising carbon footprint from turning audio into video", "url": "https://podnews.net/press-release/spotify-video-environment"}, {"title": "Music streaming is more polluting than you might think | FMT", "url": "https://www.freemalaysiatoday.com/category/leisure/2025/06/ ... [truncated]"}]

[Tool: Bash — Read Carbon Trust update and fetch white paper PDF]

CT UPDATE

LEN 13103

[gCO2] ... ews Updated calculation released on the carbon impact of online video streaming
News Updated calculation released on the carbon impact of online video streaming New white
paper published by the Carbon Trust estimates an average carbon footprint of 55gCO2e per hour
of video streaming in Europe in 2020, with viewing device as the largest determining factor.
11 June ... [truncated]

[Tool: Bash — Extract key passages from Carbon Trust white paper]

```
Exit code 1
pdftotext not found
Traceback (most recent call last):
  File "<string>", line 2, in <module>
ModuleNotFoundError: No module named 'pypdf'
Traceback (most recent call last):
  File "<stdin>", line 2, in <module>
FileNotFoundError: [Errno 2] No such file or directory: 'ct.txt'
```

[Tool: Read — /ct.pdf]

pdftoppm is not installed. Install poppler-utils (e.g. `brew install poppler` or `apt-get install poppler-utils`) to enable PDF page rendering.

[Tool: Bash — Install pypdf in scratch venv and extract white paper text]

```
WARNING: You are using pip version 21.2.4; however, version 26.0.1 is available.
You should consider upgrading via the '<scratchpad>/venv/bin/python3 -m pip install --upgrade
pip' command.
fontTools is required to fully parse the encoding of a CFF Type1 font in font dictionary
{'/BaseFont': '/CEJKOJ+MyriadPro-Regular', '/Encoding': '/WinAnsiEncoding', '/FirstChar': 67,
'/FontDescriptor': Indirect0 ... [truncated]
```

[Tool: Bash — Get Carbon Trust energy per hour, Netflix and music sources]

[Wh per hour] ... ar 2018 (IEA, 2020b)), an assumed representative mix of end-user devices (see Appendix), and the modelled average bitrate (6.40Mbps or 2.88GB/hour) based on the representative device mix. The equivalent energy consumption is a European average of 188Wh per hour of video streaming using the conventional approach (Figure 18). The European average carbon emissions per hour of video ... [truncated]

[Tool: Bash — Get Carbon Trust breakdown and Greenly music figures]

ssion (core and access) accounts for 6gCO2e/hour and 20Wh/hour (10% of total emissions and energy). Home routers account for 21gCO2e/hour and 71Wh/ hour (38% of total emissions and energy). Finally, end-user devices account for 28gCO2e/hour (25gCO2e/hour from viewing devices and 3gCO2e/hour from peripherals) and 96Wh/hour (86Wh/hour from viewing devices and 10Wh/ hour from peripherals), which make ... [truncated]

[Tool: Bash — Find Greenly music streaming per-hour figure]

[Spotify] ... have reshaped how we consume entertainment and information, offering instant access to vast content libraries from anywhere in the world. Services like Netflix, Amazon Prime Video, Disney+, Youtube, Spotify, and even Pornhub dominate global internet traffic, with billions of hours streamed each year. While this shift offers convenience and accessibility, it comes with a significant ... [truncated]

Streaming notes are drafted below. **Video streaming is much better documented than video calls. Music streaming is not:** I found no rigorous per-hour figure for it.

Draft: Video and music streaming (serves Alex and Robin)

Evidence: video (checked directly)

Source	Figure	Notes
Carbon Trust, <i>Carbon impact of video streaming</i> , white paper, June 2021 (PDF, announcement); read in full	European average, 2020: ~188 Wh and ~55 g CO₂e per hour	Breakdown: viewing devices and peripherals 96 Wh (51%) , home router 71 Wh (38%) , network 20 Wh (10%) , data centers ~1 Wh (1%) . A 50-inch TV uses ~4.5× a laptop and ~90× a phone. The authors warn the figures "are not designed to be used as representative figures for any given scenario."
Kamiya, IEA, Dec 2020 (S21)	~77 Wh per hour (2019), central estimate 36 g CO₂	Weights devices by Netflix's reported viewing mix, with 70% on TVs

Key points: - **The device dominates.** In both sources, what you watch *on* matters far more than the service or the data center. For Alex (phone, laptop, TV), the device choice is roughly a 90× spread. - **The two central estimates differ ~2.4× (77 vs. 188 Wh).** That's mainly from different assumptions about devices, routers and networks. Presenting a range seems warranted. - **For scale:** an hour of streaming at 77–188 Wh ≈ **30–70 GPT-5.5 chatbot replies** at the calculator's ~2.7 Wh.

Limitations and contrary points

- **The home router takes up 38% of the Carbon Trust figure**, but it's usually on whether or not you stream. Counting it depends on how shared energy is divided. The Carbon Trust itself says the newer "power model" (S19) better represents what *changing* your viewing actually does. Its headline figure uses the older averaging method, which suits reporting rather than individual choices.
- **So "HD vs. SD" or "stream less" savings are probably overstated** by any per-gigabyte method. This is the same issue as the camera-off claim for video calls.
- **Funding:** the Carbon Trust study was **seed-funded by Netflix** through DIMPACT, a University of Bristol-led project whose members include ITV, Sky and others. The Carbon Trust is independent, but readers like Robin should be told.
- **Operational energy only:** manufacturing the devices is excluded. The IEA notes that production is ~80% of a phone's lifetime carbon (S21), which bears on activity 9.
- **Data is from 2019–2020, and European.** The grid mix changes the carbon a lot: the Carbon Trust shows country-level variation. US figures would differ.

Evidence: music streaming (weak)

- **Greenly, "The Carbon Cost of Streaming," Feb 2025 ([page](#)).** A carbon-accounting company's "data story" written by a copywriter. It estimates Spotify's footprint, but I **couldn't locate the widely reported 1.037 g CO₂e per hour figure** on the page. It also states that Spotify's 2023 emissions report no longer includes user-device electricity. I haven't verified that against Spotify's report.
- **Boon, *The Conversation*, 25 July 2025 ([article](#)).** A University of Westminster music lecturer. It says video streaming is ~50× audio, but relays figures from others rather than measuring.
- **What seems safe to say:** audio uses far less data than video. On a phone or laptop, the device's own power probably dominates, but I have no verified per-hour figure.

Proposed source decisions: - **Carbon Trust:** use with qualifications, as the main video source, with the Netflix funding disclosed and the router allocation explained. - **IEA:** use, as a second central estimate. - **Greenly:** reject for figures, pending checks. - **The Conversation:** background only.

Question: Should I add this to [research.md](#) as section 6, recording music streaming as an evidence gap? Or try one more search for a primary music-streaming source first, such as Spotify's own emissions report?

16. Learner (RG Sureshkumar)

check spotify's emissions report

Agent

[Tool: WebSearch — query: Spotify Equity & Impact report emissions scope 3 end-user devices streaming methodology 2024 2025]

Web search results for query: "Spotify Equity & Impact report emissions scope 3 end-user devices streaming methodology 2024 2025"

Links: [{"title": "Equity & Impact Report 2024 1 Equity & Impact Report 2024", "url": "https://s29.q4cdn.com/175625835/files/doc_governance/2025/Mar/10/Spotify-Equity-Impact-Report-2024-9b1865.pdf"}, {"title": "Spotify Strikes a Chord: Big Q1 Gains and Bigger Climate Goals" ... [truncated]

[Tool: Bash — Download Spotify 2023 and 2024 reports and search for device emissions]

```
sp24.pdf: PDF document, version 1.7
sp23.pdf: PDF document, version 1.7 (zip deflate encoded)
##### sp24.pdf 63 111273
[end-use] ... examples of our key partnerships include: DIMPACT: The collaborative project brings together world-class researchers from the University of Bristol and several of the most innovative media and technology companies, striving to improve areas such as end-use-device en
... [truncated]
```

I've checked Spotify's reports directly: both the **2023 and 2024 Equity & Impact Reports** (full PDFs, 63 pages each). **Neither gives a per-hour or per-stream figure**, and since 2023 Spotify no longer counts listeners' streaming in its emissions at all.

What the reports say: - **2023 report (PDF):** - In 2023, Spotify **stopped counting "Use of sold products"** (Scope 3 Category 11). For Spotify, that category covers "the end user's device energy usage, app downloads, and data transfer energy usage." - The stated reason is a lack of industry-accepted measurement methods. - It restated 2022 emissions without these, noting that **"Previously reported 2022 end-use emissions were 103,920 tCO₂e."** - **2024 report (PDF):** - Category 11 is still excluded, "due to limited availability of relevant guidance and industry consensus." - Spotify is a member of **DIMPACT**, the same University of Bristol-led group behind the Carbon Trust's video study, working on "end-use-device energy efficiency."

What this means: - Greenly's claim that Spotify's report "no longer accounts for these emissions" is **correct**, and now verified against the primary source. - **The only primary figure is a company-wide total:** ~104,000 t CO₂e from listener devices and data transfer in 2022. Without Spotify's listening hours and method, it can't be turned into a per-hour figure, and I haven't found either. - **Music streaming stays an evidence gap.** It can be described qualitatively: much less data than video, and the device probably dominates. But I have no verified per-hour number. - **This is worth telling Robin.** A major streaming company removed the listener-side part of its footprint from its reporting because no agreed method exists. That's a documented example of the uncertainty the calculator has to communicate, and it doesn't imply bad faith either way.

Proposed decision: use the Spotify reports with qualifications, for the 2022 end-use total and the reporting change, not for a per-hour figure.

Question: Should I now add the streaming section to [research.md](#) as section 6, with sources S25 onward, recording music streaming as an evidence gap?

17. Learner (RG Sureshkumar)

yes add it to research.md

Agent

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/research.md has been updated successfully.

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/research.md has been updated successfully.

I've added the streaming notes to [research.md](#) as a draft.

- **Audience needs:** a new "6. Video and music streaming (Alex, Robin)" section. It covers:
 - the 77–188 Wh per hour video range and its component breakdown
 - the device as the dominant factor (TV ≈ 90× a phone)
 - the router-allocation caveat
 - the Netflix funding disclosure
 - music streaming recorded as an **evidence gap**, with Spotify's 2023 reporting change documented
- **Source assessments:** S25 to S28.
 - S25: Carbon Trust
 - S26: Spotify reports
 - S27: Greenly, proposed for rejection as a source of figures
 - S28: *The Conversation*, background only

Everything is still a proposal for you to confirm. Nothing is marked approved.

Next is **7. Social media use (Alex, Robin)**. I expect the evidence to be thin, much like music streaming. Likely sources are platform sustainability reports from Meta, TikTok and others, and the few academic or consultancy estimates per minute of scrolling. I'll check them against the network-energy debate (S19), and flag any that rely on per-gigabyte methods.

Question: Should I go ahead with social media?

18. Learner (RG Sureshkumar)

yes go ahead with social media

Agent

[Tool: WebSearch — query: social media carbon footprint per minute TikTok Instagram Greenspector 2021 gCO2e per minute study]

Web search results for query: "social media carbon footprint per minute TikTok Instagram Greenspector 2021 gCO2e per minute study"

Links: [{"title": "What is the environmental footprint for social media applications? 2021 Edition – Greenspector", "url": "https://blog.greenspector.com/en/social-media-2021/"}, {"title": "The Carbon Cost of Social Media – The Carbon Literacy Project", "url": "https://carbo ... [truncated]

[Tool: WebSearch — query: Meta sustainability report 2025 emissions per user data center energy per daily active user]

Web search results for query: "Meta sustainability report 2025 emissions per user data center energy per daily active user"

Links: [{"title": "Meta Platforms, Inc. – Form DEF 14A – FY2025", "url": "https://www.sec.gov/Archives/edgar/data/1326801/000162828026025532/meta-20260416.htm"}, {"title": "Meta Platforms, Inc. – Form PX14A6G – FY2025", "url": "https://www.sec.gov/Archives/edgar/data/1326801/000121 ... [truncated]

[Tool: WebSearch — query: peer-reviewed energy consumption social media apps smartphone scrolling short video measurement Wh per hour]

Web search results for query: "peer-reviewed energy consumption social media apps smartphone scrolling short video measurement Wh per hour"

Links: [{"title": "Environmental impact of top social media apps 2021 | Statista", "url": "https://www.statista.com/statistics/1177190/social-media-apps-energy-consumption-milliampere-hour-france/"}, {"title": "The Energy Cost of Social Media", "url": "https://www.t ... [truncated]

[Tool: Bash — Read Greenspector social media study and Meta report page]

```
### GREENSPECTOR SM
LEN 270

### META
LEN 6842
Shell cwd was reset to ~/Documents/College/Undergraduate School/Semesters/Fall
2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator
```

[Tool: WebFetch — <https://fortune.com/2024/12/13/tiktok-carbon-footprint-emissions-doomscrolling-greece-greenly>]

```
# TikTok Carbon Footprint Claims

**Main Claim:** According to a Greenly study cited in this December 13, 2024 Fortune article
by Sydney Lake, TikTok's annual carbon emissions are nearly equivalent to Greece's entire
national carbon footprint (approximately 50 million metric tons of CO2e).

**Per-Minute Figure:** "one minute spent on TikTok burns 2.921 grams of carbon dioxide
equivalent (CO2e)"

* ... [truncated]
```

[Tool: WebFetch — <https://blog.greenspector.com/en/social-media-2021/>]

```
The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated
tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.
```

[Tool: WebFetch — <https://sustainability.atmeta.com/2025-sustainability-report/>]

```
Based on the provided content, **no per-user, per-daily-active-user, per-hour, or per-minute
emissions or energy figures are given** for Facebook/Instagram/WhatsApp usage.

Additionally, **no total data center electricity use or total emissions figures with years are
provided** on this page.

The document mentions Meta-supported renewable energy projects adding "almost 29 gigawatts
(GW) of clean a ... [truncated]
```

[Tool: Bash — Try alternate and archived Greenspector social media URLs]

```
### https://greenspector.com/en/social-media-2021/

HTTP 404
ript> </body></html>LEN 270
### https://web.archive.org/web/2023/https://blog.greenspector.com/en/social-media-2021/

HTTP 404
```

```
r>
  </body>
</html>LEN 645
[2021] ... Wayback Machine --> DONATE Latest Show All The Wayback Machine has not archived
that URL. Click here to search for all archived pages under https://blog.greenspector.com/en ...
[truncated]
```

[Tool: WebSearch — query: greenspector.com social networks environmental footprint 2021 edition TikTok 2.63]

Web search results for query: "greenspector.com social networks environmental footprint 2021 edition TikTok 2.63"

Links: [{"title":"What is the environmental footprint for social media applications? 2021 Edition - Greenspector","url":"https://blog.greenspector.com/en/social-media-2021/"}, {"title":"What is the environmental footprint of social networking applications? 2023 Edition - Greenspector", ... [truncated]

Quick update: Greenspector's 2021 social-media study page is now offline, so I'm reading its newer 2023 edition and its methodology page instead.

[Tool: WebFetch — https://greenspector.com/en/what-is-the-environmental-footprint-of-social-networking-applications-2023/]

REDIRECT DETECTED: The URL redirects to a location that was not fetched automatically.

Original URL: https://greenspector.com/en/what-is-the-environmental-footprint-of-social-networking-applications-2023/

Redirect URL (from the server's Location header – server-supplied, not verified): https://blog.greenspector.com/en/what-is-the-environmental-footprint-of-social-networking-applications-2 ... [truncated]

[Tool: WebFetch — https://blog.greenspector.com/en/environmental-footprint-methodology/]

Greenspector's Network and Server Impact Methodology

Energy Factors

The document does not specify per-megabyte or per-GB energy factors for network or server impacts. Instead, it states that Greenspector uses "modelled data" for these components rather than direct measurements.

Device Manufacturing Impacts

Yes, embodied impacts are included. The methodology accounts for:
– "Impact of s ... [truncated]

[Tool: WebFetch — https://blog.greenspector.com/en/what-is-the-environmental-footprint-of-social-networking-applications-2023/]

Environmental Footprint of Social Networking Applications – 2023 Study

****Publication Date:**** June 21, 2023

****Apps Measured:**** Facebook, Instagram, LinkedIn, Pinterest, Reddit, Snapchat, TikTok, Twitch, Twitter, and YouTube

****Test Conditions:****

- Device: Samsung Galaxy S10, Android 10
- Network: Wi-Fi
- Screen brightness: 50%
- Minimum 3 iterations per test
- Scenario: News feed scrolling (2-s ... [truncated])

Social media notes are drafted below. **Like music, social media has no independent, peer-reviewed per-minute figure.** Everything I found traces to two commercial consultancies, and their numbers have shifted a lot between editions.

Draft: Social media use (serves Alex and Robin)

Evidence (checked directly)

Source	Figure	Notes
Greenspector, "What is the environmental footprint of social networking applications? 2023 Edition," 21 June 2023 (post)	Scrolling the feed for 1 minute on a Samsung Galaxy S10 over Wi-Fi: 0.47 g CO₂e (LinkedIn) to 0.96 g (TikTok) . Instagram and YouTube 0.87 g, Facebook 0.63 g.	10 apps, at least 3 test runs each. Network and server impacts are <i>modelled</i> , assuming servers worldwide "in the absence of information for each application." Includes device manufacturing (battery wear). Method reviewed by the life-cycle consultancy EVEA but not published in a journal (method).
Greenspector, 2021 edition	~2.63 g/min (TikTok) , average ~1.15 g/min , as quoted in search results	The original page is now offline and not archived, so I couldn't verify these figures. If the numbers are right, TikTok's figure fell ~2.7× between editions, probably from method and device changes. Either way, that shows how unstable these estimates are.
Greenly, reported by Sydney Lake, <i>Fortune</i> , 13 Dec 2024 (article)	2.921 g CO₂e per minute on TikTok , and total TikTok emissions said to be "nearly equivalent to Greece"	TikTok disputed it , saying ByteDance's total 2023 emissions were "less than 20% of the estimated emissions by Greenly."
Meta, <i>2025 Sustainability Report</i> page (page)	No per-user, per-hour or per-minute figure	I checked only the summary page, not the full report document.

What the evidence suggests

- **Per minute of scrolling, estimates range from ~0.5 to ~3 g CO₂e**, a roughly 6× spread across sources and years for the same kind of activity.
- **The platform matters less than the source's method.** Within one Greenspector edition, apps differ only ~2×. Between sources and editions, the same app (TikTok) differs ~3×.
- **Video-heavy feeds** (TikTok, YouTube, Instagram) come out at the higher end, consistent with the device doing more work and more data being transferred.

Limitations and contrary points

- **The sources are commercial**, from companies that sell "digital sobriety" or carbon-accounting services. **Neither is peer reviewed.**
- **A platform directly disputes one of them**, and neither side's claim can be independently checked.
- **Network and server parts are modelled**, probably per data volume, although Greenspector doesn't publish its factors. That runs into the same objection as before (S19).
- **All tests used one Android phone**, scrolling for 1 minute. Laptop or TV use (relevant to Alex and Robin) isn't covered.
- **No platform (Meta, TikTok) publishes per-user or per-hour figures** that I found.

Assumptions still to check

- **Whether social media is best modelled as "video streaming on a phone,"** using the device-dominated streaming evidence (S21, S25) instead of these consultancy figures. That's a methodology choice for you.
- **Whether Meta's full report has a per-user intensity metric.** Not checked.

Proposed source decisions: - **Greenspector 2023:** use with strong qualifications, or reject. Your call. - **Greenspector 2021:** reject, as unverifiable. - **Greenly via Fortune:** use only as a documented, disputed claim. - **Meta:** record as "no disclosure found."

Question: Should I add this to [research.md](#) as section 7, or check Meta's full sustainability report first?

19. Learner (RG Sureshkumar)

check metas full sustainability report first

Agent

[Tool: Bash — Find Meta full sustainability report PDF link]

```
href="http://sustainability.atmeta.com/asset/2025-sustainability-report/"
href="https://sustainability.atmeta.com/2025-sustainability-report/"
href="https://sustainability.atmeta.com/asset/2025-environmental-data-index/"
href="https://sustainability.atmeta.com/asset/2025-independent-accountants-review-report/"
href="https://sustainability.atmeta.com/asset/2025-sustainability-report#page=48"
href=" ... [truncated]
```

[Tool: Bash — Download Meta report and data index and search for per-user metrics]

```
meta_rep.pdf: PDF document, version 1.7 (zip deflate encoded)
meta_idx.pdf: PDF document, version 1.7 (zip deflate encoded)
##### meta_rep.pdf 74 114376
[intensity] ... native to conventional diesel fuel. 32% in office energy consumption 50% in
waste generated by office operations 35% in employee commute emissions 40% in embodied carbon
of building materials and furniture 54% in carbon intensity o ... [truncated]
```

[Tool: Bash — Check definitions of daily active person and use of sold products]

```
meta_idx.pdf [daily active] ... - - 160 3,154 Environmental Footprint 1,2,3,4,5 GHG Intensity
Market-Based Scope 1 and 2 Emissions (in metric tons CO2 e/unit of key performance indicators)
2020 2021 2022 2023 2024 GHG intensity per daily active person 0.00002 0.00002 0.00002
0.000016 0.000015 GHG intensity per million USD of revenue - 0.49 0.58 0.43 0.30 GHG intensity
per MWh - 0.0061 0.0058 0.003 ... [truncated]
```

I've checked **Meta's full 2025 Sustainability Report** (74 pages) and its **2025 Environmental Data Index** (19 pages). Unlike the summary page, **the data index does include per-user figures**. They cover only Meta's side, not the user's phone or network.

What Meta discloses ([data index PDF](#); [report PDF](#)):

Metric (2024)	Meta's figure	In everyday units
Electricity per daily active person	0.0055 MWh	~5.5 kWh per person per year, ≈ 15 Wh per day
Scope 1+2 carbon per daily active person (market-based)	0.000015 t CO ₂ e	~15 g CO ₂ e per person per year
Total electricity	18.4 TWh (18.1 TWh data centers)	Up from 7.2 TWh in 2020

- **I cross-checked the arithmetic.** Total electricity divided by the intensity implies ~3.35 billion daily active people. Market-based Scope 1+2 (47,468 + 1,358 t) divided by that gives ~15 g, which matches.
- **For scale,** 15 Wh per day is about 6 GPT-5.5 chatbot replies, or roughly 5–10 minutes of video streaming at the S21/S25 figures.

Why it can't be used directly as "energy per hour of social media": - It's a whole-company average, spread over everyone who uses any Meta app at least once a day. It isn't tied to time spent, and it includes Meta's AI training, advertising systems and other workloads, not just serving feeds. - **It excludes the user's own device and the network**, which the streaming evidence (S21, S25) suggests are the larger share. - **The carbon figure is market-based.** Meta matches 100% of its electricity with renewable purchases, so the carbon looks very low. Meta's own location-based total is much higher: 15.6 million t CO₂e across all scopes in 2024. This is the same market-vs-location issue as Google (S14, S17). - **"Use of sold products" (17,521 t)** refers only to Meta's own hardware, such as Quest headsets, **not people's phones**. - **I couldn't interpret the water metric.** The per-person water withdrawal of 0.001683 is labelled "liters," but that doesn't square with plausible totals. I'd propose not using it.

What this adds: - A **company-disclosed, independently reviewed data point** for the data-center share of social media. An independent accountants' review report is linked alongside. - **It supports the pattern seen elsewhere:** server-side energy per user is small, and the user's device and network probably dominate.

Proposed decision: use with qualifications, as the data-center share only, labelled as a per-person yearly average and not per hour, with the market-based caveat.

Question: Should I add the social media section to [research.md](#) as section 7, including the Meta figures, as sources S29 onward?

20. Learner (RG Sureshkumar)

yes add it to research.md

Agent

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

[Tool: Bash — Get the Fortune article headline]

```
<title>Doomscrolling has reportedly made TikTok's annual carbon footprint almost the same as the entire country of Greece | Fortune
og:title" content="Doomscrolling has reportedly made TikTok's annual carbon footprint almost the same as the entire country of Greece | Fortune
<title>Down Arrow Button Icon
```

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

I've added the social media notes to [research.md](#) as a draft.

- **Audience needs:** a new "7. Social media use (Alex, Robin)" section. It covers:
 - the ~0.5–3 g CO₂e per minute range across consultancy estimates, and how unstable those estimates are
 - TikTok's dispute of Greenly's figure
 - Meta's disclosed data-center figures: ~5.5 kWh per daily active person per year, ≈ 15 Wh per day
 - the modelling choice, recorded as yours to decide
- **Source assessments:** S29 to S32.
 - S29: Greenspector 2023
 - S30: Greenspector 2021, proposed for rejection as unverifiable
 - S31: Greenly via *Fortune*, as a disputed claim
 - S32: Meta's full report and data index

On S32: I didn't read the independent accountants' review that Meta publishes alongside its figures, and the entry says so.

Everything is still a proposal for you to confirm. Nothing is marked approved.

Next is **8. Gaming (Jordan)**, covering desktop PC and console. The calculator already has an "hour on a PS5" comparison (0.08 kg CO₂ per hour), so I'd first trace where that figure comes from. Then I'd look for measured power draw for gaming PCs and consoles, and cloud-gaming estimates.

Question: Should I go ahead with gaming?

21. Learner (RG Sureshkumar)

yes go ahead with gaming

Agent

[Tool: Bash — Find source of PS5 figure in calculator]

```
assets/js/app.js:23:      try { AIPF[name](root); } catch (e) { console.error('AIPF ' + name +
' failed', e); }
index.html:406:      An hour on a <a href="https://www.playstation.com/en-
us/legal/ecodesign/" target="_blank" rel="noopener">PS5 (~200 W)</a>;
index.html:422:      <strong>How the appliance figures are derived.</strong> The coffee,
PS5, microwave, shower,
assets/js/data.js:103:    { ... [truncated]
```

[Tool: Bash — Read calculator methodology around PS5 figure]

```
a new smartphone ≈ 70 kg (<a href="https://www.apple.com/environment/" target="_blank"
rel="noopener">Apple</a>);
    and a pair of jeans ≈ 33 kg (<a href="https://www.levistrauss.com/wp-
content/uploads/2015/03/Full-LCA-Results-Deck-FINAL.pdf" target="_blank"
rel="noopener">Levi's</a>).
    </p>
    <p>
        <strong>Everyday things.</strong> A cup of coffee: ~0.21 kg CO<sub>2</sub>e and ...
    [truncated]
```

[Tool: Bash — Read Sony PlayStation ecodesign power page]

```
LEN 2044
[W] ... and/or trade dress, trademarks, artwork and associated imagery are trademarks and/or
copyright material of their respective owners. All rights reserved. More info --> --> -->
Country / Region: United States--> --> United States Legal Privacy policy Website terms of use
Site map Do not share my personal information Back to PlayStation Age restricted 404 Enter
your date of birth. M ... [truncated]
```

[Tool: WebSearch — query: Mills Bourassa Rainer "Toward greener gaming" gaming computers energy use Energy Efficiency journal kWh per year]

Web search results for query: "Mills Bourassa Rainer "Toward greener gaming" gaming computers energy use Energy Efficiency journal kWh per year"

Links: [{"title": "Toward Greener Gaming: Estimating National Energy Use and Energy Efficiency Potential | The Computer Games Journal | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s40869-019-00084-2"}, {"title": "Publications - Gr ... [truncated]"}]

[Tool: WebSearch — query: cloud gaming energy consumption carbon per hour study peer-reviewed Marsden Lancaster versus local console]

Web search results for query: "cloud gaming energy consumption carbon per hour study peer-reviewed Marsden Lancaster versus local console"

Links: [{"title": "Your Screen, Your Footprint. The real environmental cost of... | by Meghannbf | Medium", "url": "https://medium.com/@meghannbf/your-screen-your-footprint-dd9cd535685d"}, {"title": "Hot Games: Towards a Holistic Assessment of the Planet Warming Emis ... [truncated]"}]

[Tool: WebSearch — query: PlayStation 5 power consumption watts official Sony ecodesign CFI-2000 gameplay W]

Web search results for query: "PlayStation 5 power consumption watts official Sony ecodesign CFI-2000 gameplay W"

Links: [{"title": "PS5 Power Consumption: Complete Electricity Usage Guide 2025", "url": "https://solartechonline.com/blog/ps5-electricity-usage-power-consumption-guide/"}, {"title": "PS5 Power Draw: How Much Electricity Does It Use? [Real Costs] - GamerHorizon.blog", "url": "https://gamerho ... [truncated]"}]

[Tool: Bash — Read Hot Games abstract and Mills et al. article]

Hot Games: Towards a Holistic Assessment of the Planet Warming Emissions of Video Games based on 2024-2025 Data
['Hazas, Mike', 'Dalli, Kevin C.', 'Menon, Arjun', 'Abraham, Ben', 'Nordgren, Ossian']
['2026/08/19']

Abstract: Following on recent reports on specific platforms or companies, this paper provides an assessment of the global impact of the production and use of video games. It draws together ...
[truncated]

[Tool: Bash — Try Sony ecodesign page variants and archive]

<persisted-output>

Output too large (341.5KB). Full output saved to: <local session log path>

Preview (first 2KB):

https://www.playstation.com/en-gb/legal/ecodesign/

HTTP 200

<script defer crossorigin src="https://static.playstation.com/wca/v2/js/ui-

```
components.61c619ec95e5f5b047a9.js" type="application/javascript"></script><script defer  
crossorigin src="https://static.playstation.com/wca ... [truncated]
```

Quick update: the calculator's own PS5 source link (the US Sony page) now returns a 404. I found Sony's UK "Energy Efficiency" page and I'm extracting the PS5 power figures from it.

[Tool: Bash — Extract PS5 power figures from Sony UK energy page]

```
51735  
[PlayStation 5] ... Energy Efficiency | Legal | PlayStation.com Ecodesign Parts and repair  
resources Select PlayStation 5 console spare parts are now available for sale to independent  
repairers. Spare parts may be new or refurbished, meeting original manufacturer standards.  
These can be purchased from the following Sony official partners: EET ASW0 Repair Guides can  
be accessed here The belo ... [truncated]
```

[Tool: Bash — Get full PS5 power table and LBL cloud gaming page]

```
PlayStation®5 model number: CFI-21 (digital) HD Active gaming (PS5 games: three game average):  
215.7 W Active gaming (PS4 game: Marvel's Spider-Man): 105.1 W Active gaming (PS4 game:  
Battlefield 4): 88.4 W Home menu user interface: 43.4 W UHD Active gaming (PS5 games: three  
game average): 217.19 W Home menu user interface: 44.53 W Low power Rest*: 0.30 W Rest (Supply  
Power to USB Ports): 4.65 W Re ... [truncated]
```

[Tool: Bash — Read LBL Green Gaming pages and Mills citation metadata]

```
### /  
LEN 3511  
[ W] ... S. households . The average U.S. gamer is 35 years old. The number of gamers is  
rising, as is the amount of time spent gaming. Gaming is particularly popular in California,  
where an estimated 15 million non-battery-powered gaming devices are located. We have  
estimated the associated global energy cost at $5 billion each year for higher-end gaming PCs  
alone, and rising. Our ... [truncated]
```

[Tool: WebFetch — <https://link.springer.com/article/10.1007/s40869-019-00084-2>]

```
REDIRECT DETECTED: The URL redirects to a location that was not fetched automatically.  
  
Original URL: https://link.springer.com/article/10.1007/s40869-019-00084-2  
Redirect URL (from the server's Location header – server-supplied, not verified):  
https://idp.springer.com/authorize?  
response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Fart  
icle%2F10.1007%2Fs ... [truncated]
```

[Tool: Bash — Read LBL publication page for Mills et al.]

```
LEN 60  
Shell cwd was reset to ~/Documents/College/Undergraduate School/Semesters/Fall  
2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator
```

[Tool: WebSearch — query: [escholarship "Toward Greener Gaming" Mills Bourassa Rainer Mai Shehabi pdf](#)]

Web search results for query: "escholarship "Toward Greener Gaming" Mills Bourassa Rainer Mai Shehabi pdf"

```
Links: [{"title": "Lawrence Berkeley National Laboratory LBL Publications  
Title", "url": "https://escholarship.org/content/qt3sg2j75k/qt3sg2j75k.pdf?v=lg"},  
{"title": "Publications – Green Gaming", "url": "https://greengaming.lbl.gov/publications"},  
{"title": "Evan Mills, Ph.D – Energy Systems Analys ... [truncated]}
```

Quick update: I found an open copy of the main gaming-energy study (Mills et al. 2019, Berkeley Lab) and I'm extracting its measured power figures now.

[Tool: Bash — Extract measured gaming power figures from Mills et al.]

```
Exit code 1  
mills.pdf: HTML document text, ASCII text
```

[Tool: WebFetch — https://escholarship.org/content/qt3sg2j75k/qt3sg2j75k.pdf?v=lg]

I appreciate your detailed request, but I need to be transparent: the content you've provided appears to be a PDF file rendered as raw binary/encoded data, which is not readable as plain text.

The document shows PDF structural elements (object references, stream encoding, font definitions) but the actual article content is compressed/encrypted and cannot be extracted into readable text form throu ... [truncated]

[Tool: Bash — Extract text from the saved Mills et al. PDF]

```
259 647413  
[watts] ... dustry rapidly matured, and Bitcoin mining has attained extraordinary scale,  
equivalent to 3.2 billion high-end GPU's" (Xie et al. 2018). The mid-2018 state-of-the-art 16  
nm ASIC design "Bitfury attains better than 0.07 watts per GH/s energy efficiencies, an  
improvement over 100× over the first 130-nm [ASIC] miners, and over 8,000× over GPUs." Despite  
tremendous gains in com ... [truncated]
```

[Tool: Bash — Fetch Mills et al. abstract from Semantic Scholar API]

```
{  
  "paperId": "cd0f25d5ca4c5d6ef767efce38bdb1586301dcfb",  
  "title": "Toward Greener Gaming: Estimating National Energy Use and Energy Efficiency  
Potential",  
  "venue": "THE COMPUTER GAMES JOURNAL",  
  "year": 2019,  
  "openAccessPdf": {  
    "url": "",  
    "status": null,  
    "license": null,  
    "disclaimer": "Notice: The following paper fields have been elided by the pub ...  
[truncated]
```

[Tool: Browser navigate — https://link.springer.com/article/10.1007/s40869-019-00084-2]

navigated to <https://link.springer.com/article/10.1007/s40869-019-00084-2>

Tab Context:

– Executed on tabId: seed

– Available tabs:

- tabId seed: "Toward Greener Gaming: Estimating National Energy Use and Energy Efficiency Potential | The Computer Games Journal | Springer Nature Link" ("<https://link.springer.com/article/10.1007/s40869-019-00084-2>")

[Tool: Browser get_page_text —]

Title: Toward Greener Gaming: Estimating National Energy Use and Energy Efficiency Potential | The Computer Games Journal | Springer Nature Link

URL: <https://link.springer.com/article/10.1007/s40869-019-00084-2>

Source element: <article>

The Computer Games Journal Article

Toward Greener Gaming: Estimating National Energy Use and Energy Efficiency Potential Research

Published: 30 October 2019

V ... [truncated]

[Tool: Browser navigate — <https://link.springer.com/article/10.1007/s12053-015-9371-1>]

navigated to <https://link.springer.com/article/10.1007/s12053-015-9371-1>

Tab Context:

– Executed on tabId: seed

– Available tabs:

- tabId seed: "Taming the energy use of gaming computers | Energy Efficiency | Springer Nature Link" ("<https://link.springer.com/article/10.1007/s12053-015-9371-1>")

[Tool: Browser find — Abstract]

Found 1 match(es) for "Abstract":

– heading "Abstract" [ref_90]

Tab Context:

– Executed on tabId: seed

– Available tabs:

- tabId seed: "Taming the energy use of gaming computers | Energy Efficiency | Springer Nature Link" ("<https://link.springer.com/article/10.1007/s12053-015-9371-1>")

Quick update: I'm reading the abstract of a second Berkeley Lab study on gaming PCs (Mills & Mills 2015) to get a per-machine energy figure. Then I'll write up the gaming notes.

[Tool: Browser javascript_tool — javascript_exec]

"Abstract\n\nOne billion people around the world engage in some form of digital gaming. Gaming is the most energy-intensive use of personal computers, and the high-performance "racecar" systems built expressly for gaming are the fastest growing type of gaming platform. Large

performance-normalized variations in nameplate power ratings for gaming computer components available on today's market indi ... [truncated]

Gaming notes are drafted below. First, a problem I found in the existing calculator: **its PS5 citation link is broken**. playstation.com/en-us/legal/ecodesign/ now returns a 404. Sony's current figures are on its UK page, and they're close to the calculator's ~200 W.

Draft: Gaming (serves Jordan)

Evidence (checked directly)

| Source | Figure | Notes |
|--|---|--|
| Sony Interactive Entertainment, PlayStation "Energy Efficiency" page, UK Games Console Voluntary Agreement statement (page) | Current PS5 (CFI-21/CFI-71): ~214–219 W playing PS5 games (average of 3 titles, HD or UHD). 88–110 W playing PS4 games . ~43–47 W on the home menu. 0.3 W in low-power rest. | Manufacturer self-test dated 02/05/2026 on named titles (Death Stranding 2, Demon's Souls, COD Black Ops 7). The calculator's ~200 W is consistent , but its link needs updating. |
| Mills, Bourassa, Rainer, Mai, Shehabi & Mills, "Toward Greener Gaming," <i>The Computer Games Journal</i> 8:157–178, 2019 (DOI ; abstract read) | 26 gaming systems measured. Energy varied as much by which of 37 games was played as by hardware . Cloud gaming uses "markedly higher" energy than local. US gaming $\approx$ 34 TWh/yr, 2.4% of home electricity . | Berkeley Lab, funded by the California Energy Commission, no declared conflicts. Paywalled, so I read only the abstract. Hardware from 2017–18. |
| Mills & Mills, "Taming the energy use of gaming computers," <i>Energy Efficiency</i> 9:321–338, 2016 (DOI ; abstract read) | A typical gaming PC including its display $\approx$ 1,400 kWh per year . Intensive users "could easily" use double. Measured peak power ran ~50% below the nameplate rating. | Based on 5 measured PCs with circa-2015 components. A yearly figure, not per hour, and I can't see the assumed hours of use in the abstract. |
| Berkeley Lab Green Gaming, "Cloud Gaming" page (page) | Cloud gaming needs significantly more energy than similar home hardware, up to 3x in the most extreme cases , because of data-center cooling plus network energy | The same research group, in summary form |
| Hazas, Dalli, Menon, Abraham & Nordgren, "Hot Games," arXiv, Aug 2026 (arXiv) | A global emissions assessment of video games using 2024–25 data, updating their 2020 and 2022 figures | Preprint. Abstract only, which has no per-hour figure. Could be read further if needed. |

For scale: - One hour of PS5 gaming at ~216 Wh $\approx$ **80 GPT-5.5 chatbot replies** at the calculator's figure. - That's about the same as an hour of streaming at the Carbon Trust's 188 Wh. - Jordan's desktop PC could be well above a console, but I don't have a verified per-hour figure for current PCs.

Limitations and contrary points

- **Gaming PC figures are old.** The best per-machine number dates from 2016, with 2015 hardware. Modern GPUs are more efficient per frame but often draw more power in total. That direction isn't evidenced here.
- **Which game you play matters as much as the hardware** (Mills 2019). So one figure per platform hides a wide range. Sony's own figures show the same thing: PS4 titles run at about half the power of PS5 titles.

- **Sony's figures are manufacturer self-tests.** They're given under a voluntary agreement, not independent testing.
- **The display is excluded from console figures** but included in the gaming-PC figure, so they aren't directly comparable.
- **Only operational energy is covered.** Manufacturing the hardware is left to activity 9.

Assumptions still to check

- **Current measured power draw of a desktop gaming PC** during play, for Jordan. That needs a recent independent source, such as hardware-review measurements or newer Berkeley Lab work.
- **How many hours per week Jordan-like users game.** No evidence gathered.

Proposed source decisions: - **Sony:** use with qualifications, replacing the calculator's dead link. - **Mills 2019 and Mills & Mills 2016:** use with qualifications, dated and abstract-level. - **Berkeley Lab cloud page:** use with qualifications. - **Hot Games:** background, pending a fuller read.

Question: Should I add this to [research.md](#) as section 8, or first look for a recent independent measurement of desktop gaming PC power?

22. Learner (RG Sureshkumar)

look for a recent independent measurement of desktop gaming pc power

Agent

[Tool: WebSearch — query: measured wall power gaming PC system watts during gameplay study 2024 2025 RTX 4070 4090 whole system energy per hour]

Web search results for query: "measured wall power gaming PC system watts during gameplay study 2024 2025 RTX 4070 4090 whole system energy per hour"

Links: [{"title":"How Much Electricity Does A Gaming PC Use? [2025 Guide]","url":"https://solartechonline.com/blog/how-much-electricity-does-gaming-pc-use/"}], [{"title":"How to Determine the Power Supply Requirements for Installing an NVIDIA RTX 4090 ... [truncated]

[Tool: WebSearch — query: "gaming" desktop PC energy consumption measurement peer-reviewed 2023 2024 2025 kWh per hour of play arXiv]

Web search results for query: ""gaming" desktop PC energy consumption measurement peer-reviewed 2023 2024 2025 kWh per hour of play arXiv"

Links: [{"title":"A Comparative Analysis of Energy Consumption Between the Widespread Unity and Unreal Video Game Engines","url":"https://arxiv.org/html/2402.06346v2"}, {"title":"From Clicks to Carbon: The Environmental Toll of Recommender Systems","url":"https ... [truncated]

[Tool: Bash — Read Unity vs Unreal energy paper details]

A Comparative Analysis of Energy Consumption Between The Widespread Unreal and Unity Video Game Engines
['Pérez, Carlos', 'Verón, Javier', 'Pérez, Francisca', 'Moraga, M Ángeles', 'Calero, Coral', 'Cetina, Carlos'] ['2024/02/09']
Comments: None
Journal: None
LEN 63787
[358] ... between 230 TWh (just considering PC gamers) and 347 TWh (including gaming consoles). The estimation was made by measurin ... [truncated]

[Tool: Bash — Search Hot Games paper for per-hour device power assumptions]

LEN 35613
[watt] ... , however, and a November 2025 scrape (Abraham, 2026) provides a basis for an estimation – showing a combined total of 5.57b player hours on Steam for the prior 30 days, across 12k games. Applying a weighted average power consumption figure (305.1 watts) from the Sustainable Games Alliance’s Steam hardware model, which combines Steam’s Hardware survey with a database of the ... [truncated]

[Tool: Bash — Find hardware and reference for the Baldur's Gate measurement]

[\[4\]]statista.com/topics/1680/gaming/#topic0verview , accessed: 2023–11–12. [3] How many gamers are there? (new 2023 statistics), https://explodingtopics.com/blog/number-of-gamers , accessed: 2023–11–12. [4] J. Clement, Most played games on steam as of september 2023, by average number of concurrent players, https://www.statista.com/statistics/1179973/steam-games-peak-concurrent-players/ ... [truncated]

I searched for a recent independent measurement. **There's no recent, independent, whole-system measurement of typical desktop gaming PCs** comparable to Berkeley Lab's older work. I found one small lab measurement and one modelled estimate, and each has a different weakness.

| Source | Figure | Type | Weakness |
|--|---|--|---|
| Pérez, Verón, Pérez, Moraga, Calero & Cetina, "A Comparative Analysis of Energy Consumption Between the Widespread Unreal and Unity Video Game Engines," arXiv, Feb 2024 (arXiv) | 358.6 W average while playing <i>Baldur's Gate 3</i> | Measured with a hardware meter (their "EET" device) on one test PC | One game on one PC. Background power is subtracted and the monitor excluded, so a real player draws somewhat more. Mentioned in passing in the introduction, and I couldn't find the test PC's specification. Preprint. |
| Hazas, Dalli, Menon, Abraham & Nordgren, "Hot Games," arXiv, Aug 2026 (arXiv); reading more than the abstract this time | 305.1 W weighted average across PCs on Steam | Modelled: Steam's hardware survey combined with component power ratings (TDPs), from the Sustainable Games Alliance | Based on rated power, not measured. Mills & Mills (2016) found measured peak power ran ~50% <i>below</i> nameplate for complete systems, so this could be high. Excludes the monitor. Preprint. |

What this suggests: - ~300–360 W for the PC itself while gaming, plus a monitor. That's roughly 1.5× a PS5 (~216 W), and about 110–130 GPT-5.5 chatbot replies per hour at the calculator's figure. - **The two sources point the same way from different directions**, one measured and narrow, one modelled and broad. But neither is strong alone, so a range with clear labels seems warranted. - **High-end systems could be well**

above this. Commercial blogs claim 400–900 W for systems with powerful graphics cards, but they cite no measurements, so **I propose rejecting them.** The Mills & Mills finding that intensive users "could easily" use double the typical figure is the evidence-based version of that point.

Another detail from "Hot Games" relevant to Jordan and to cloud gaming: it raises the Carbon Trust's streaming delivery figure by 50% for cloud gaming, because cloud gaming uses more bandwidth (10–40 Mbps vs 2–16 Mbps, citing Marsden et al. 2020). That is a per-data adjustment of the kind S19 criticises, so I'd treat it with caution.

Proposed decisions: - **Pérez et al. 2024:** use with qualifications, as one measured data point. - **Hot Games 305 W:** use with qualifications, as a modelled estimate that may run high. - **Commercial wattage blogs:** reject.

Question: Should I add the gaming section to [research.md](#) as section 8, including these two sources and the note about the broken PS5 link, as sources S33 onward?

23. Learner (RG Sureshkumar)

yes add it to research.md

Agent

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

I've added the gaming notes to [research.md](#) as a draft.

- **Audience needs:** a new "8. Gaming (Jordan)" section. It covers:
 - the PS5 at ~214–219 W
 - desktop gaming PCs at ~300–360 W before the monitor
 - the older ~1,400 kWh per year figure
 - variation by game
 - cloud gaming at up to 3× local
 - the **broken PS5 link in the current calculator**
- **Source assessments:** S33 to S38.
 - S33: Sony
 - S34: Mills et al. 2019
 - S35: Mills & Mills 2016

- S36: Pérez et al. 2024
- S37: Hot Games
- S38: Berkeley Lab's cloud page

Two notes: - I read only the abstracts of S34 and S35, because both are paywalled. The entries say so. - **The broken link is recorded, not fixed.** Changing [index.html](#) belongs to implementation, so it's in the notes as something to fix then.

Everything is still a proposal for you to confirm. Nothing is marked approved.

Next is **9. Device manufacturing (embodied impacts), which concerns all three profiles.** Several sections have pointed here. The IEA says making a phone is ~80% of its lifetime carbon, and the calculator already cites Apple for a phone (~70 kg) and has a laptop (~250 kg) and TV (~350 kg). I'd check those figures against current manufacturer reports (Apple, Dell, and TV makers) and independent life-cycle studies. I'd also look at how the calculator could spread manufacturing impacts over hours of use.

Question: Should I go ahead with device manufacturing?

24. Learner (RG Sureshkumar)

yes go ahead with device manufacturing

Agent

[Tool: Bash — Find device figures and sources in calculator]

```
385:      a <a href="https://www.epa.gov/sites/default/files/2018-02/documents/lca_tv.pdf"
target="_blank" rel="noopener">new flat-screen TV ≈ 350 kg</a>;
386:      a <a href="https://devera.ai/benchmarks/carbon-footprint-of-a-laptop"
target="_blank" rel="noopener">new laptop ≈ 250 kg</a>;
400:      a new smartphone ≈ 70 kg (<a href="https://www.apple.com/environment/"
target="_blank" rel="n ... [truncated]
```

[Tool: Bash — Check calculator's device source links respond]

```
200 application/pdf https://www.epa.gov/sites/default/files/2018-02/documents/lca_tv.pdf
200 text/html; charset=utf-8 https://devera.ai/benchmarks/carbon-footprint-of-a-laptop
200 text/html; charset=utf-8 https://www.apple.com/environment/
200 text/html; charset=utf-8 https://www.compareandrecycle.co.uk/blog/world-water-week-how-
much-water-does-your-smartphone-use
```

[Tool: WebSearch — query: Apple iPhone 17 Product Environmental Report carbon footprint kg CO2e life cycle production percentage]

```
Web search results for query: "Apple iPhone 17 Product Environmental Report carbon footprint
kg CO2e life cycle production percentage"
```

```
Links: [{"title": "iPhone 17e | Product Environmental
```

```
Report","url":"https://www.apple.com/environment/pdf/products/iphone/iPhone_17e_PER_Mar2026.pdf"}, {"title": "iPhone Sustainability Report: What You Need to Know", "url": "https://www.compareandrecycle.co.uk/blog/ip ... [truncated]
```

[Tool: WebSearch — query: Dell laptop product carbon footprint Latitude kg CO2e PCF manufacturing share use phase 2025]

Web search results for query: "Dell laptop product carbon footprint Latitude kg CO2e PCF manufacturing share use phase 2025"

```
Links: [{"title": "Carbon Footprint of a Dell Latitude 6430", "url": "https://i.dell.com/sites/csdocuments/Corporate_corp-Comm_Documents/en/WP-E6430.pdf"}, {"title": "Carbon Footprint of a Dell Latitude 6530", "url": "https://i.dell.com/sites/csdocuments/Corporate_corp-Comm_Docume ... [truncated]
```

[Tool: Bash — Extract iPhone 17 life-cycle carbon figures]

```
ip17.pdf: PDF document, version 1.5 (zip deflate encoded)
5 15326
[kg CO2e] ... photo ID (local law may require saving this information). Additional terms from Apple or Apple's trade-in partners may apply. Carbon Footprint Endnotes Configuration
Greenhouse gas emissions 256GB 55 kg CO2e 512GB 61 kg CO2e Power consumption for iPhone 17
Mode 100V 115V 230V Power adapter, no-load 0.04W 0.04W 0.04W Po ... [truncated]
```

[Tool: Bash — Check what Apple means by net GHG emissions]

```
[Net GHG] ... eets our internal standards. Since recycled material typically emits less carbon than newly mined materials over its lifetime, we look to postindustrial and postconsumer sources for waste streams that can be recovered and used in new products. 55 kg Net GHG emissions (CO2e)4 iPhone 17 30% recycled content, 1 including: 85% recycled aluminum in the enclosure 100% recycled cobalt in th ... [truncated]
```

[Tool: WebSearch — query: MacBook Air M4 Product Environmental Report PDF apple.com/environment/pdf/products/notebooks]

Web search results for query: "MacBook Air M4 Product Environmental Report PDF apple.com/environment/pdf/products/notebooks"

```
Links: [{"title": "M4 MacBook Air Product Environmental Report", "url": "https://www.apple.com/environment/pdf/products/notebooks/M4_MacBook_Air_PER_March2025.pdf"}, {"title": "MacBook Air 15\" M2 Environmental Report | PDF | Recycling | Life Cycle Assessment", "url": "https://www ... [truncated]
```

[Tool: WebSearch — query: Clément Jacquemotte Hilty 2020 sources of variation life cycle assessments smartphones tablet computers Journal of Cleaner Production]

Web search results for query: "Clément Jacquemotte Hilty 2020 sources of variation life cycle assessments smartphones tablet computers Journal of Cleaner Production"

```
Links: [{"title": "Sources of variation in life cycle assessments of smartphones and tablet
```

```
computers","url":"https://www.zora.uzh.ch/entities/publication/d79e8840-3a33-4dcc-8593-9f96a2884123"}, {"title":"Sources of variation in life c ... [truncated]
```

[Tool: WebSearch — query: television life cycle assessment carbon footprint manufacturing kg CO2e 55 inch peer-reviewed OR Samsung product carbon footprint TV]

Web search results for query: "television life cycle assessment carbon footprint manufacturing kg CO2e 55 inch peer-reviewed OR Samsung product carbon footprint TV"

Links: [{"title":"The carbon footprint of watching television, comparing digital terrestrial television with video-on-demand","url":"https://www.researchgate.net/publication/241184771_The_carbon_footprint_of_watching_television_compar ... [truncated]

[Tool: Bash — Extract MacBook Air and EPA TV life-cycle figures]

```
mba.pdf: PDF document, version 1.4, 12 pages
tv.pdf: PDF document, version 1.6 (zip deflate encoded)
clem.pdf: HTML document text, ASCII text
### MBA 12
[kg CO2e] ... footprint We've also calculated the product carbon footprint for different
configurations: Greenhouse gas emissions MacBook Air 15-inch with M4 (512GB SSD) Apple
emissions from utility-purchased electricity (scope 2) 0 kg CO2e Li ... [truncated]
```

[Tool: Bash — Check whether cited TV and laptop sources state the calculator's figures]

```
12021
350 0
per unit 0
per TV 0
weight 6
... Parameter Energy for production Average (weighted) production energy Process Emissions
to air (kg) Emissions to water (kg) Total energy demand (MJ/kg) Material inputs (kg) GWP, 100
yr (kg CO2- equi
... sources Palladium 14,588 4,898 183,737 5,859 9,297 Ecoinvent, 2012. Palladium at
regional storage. European mix, from global sources Precious Metal ... [truncated]
```

[Tool: Bash — Extract findings from Clément et al. 2020 review]

```
Exit code 1
clem.pdf: HTML document text, Unicode text, UTF-8 text, with very long lines (2504)
```

[Tool: WebSearch — query: Samsung OR LG TV product carbon footprint certified kg CO2e 55-inch 65-inch life cycle manufacturing use phase]

Web search results for query: "Samsung OR LG TV product carbon footprint certified kg CO2e 55-inch 65-inch life cycle manufacturing use phase"

Links: [{"title":"LG Earns Multiple Eco-Certifications for Its 2026 TV Lineup – Residential


```
Systems","url":"https://www.residentialsystems.com/news/lg-earns-multiple-eco-certifications-for-its-2026-tv-lineup"}, {"title":"Multiple Eco-Certifications for 2026 ... [truncated]
```

[Tool: Bash — Extract Samsung QLED TV environmental report figures]

```
qled.pdf: PDF document, version 1.5, 11 pages
11 14815
[kg] ... h improving product energy efficiency, we are trying to improve the environmental
impact at the use stage. Sourcing Recycling Production Use Distribution QLED 75Q60C Life Cycle
carbon emissions : 982 kg CO2eq. [2] 20% 40% 60% 100%10% Disposal 1.0 % Use 26.5 % Production
70.4% Distribution 2.0% 80% * The figure above calculates the env ... [truncated]
```

Device manufacturing notes are drafted below. **Checking the calculator's existing device citations turned up two problems:**

1. **The TV figure (~350 kg) isn't supported by the source it cites.** The linked EPA document is a table of emission factors *per kilogram of material* for a 32-inch LCD TV model. It contains no per-TV total and no "350."
2. **The phone and laptop figures are out of date.** - Apple's current iPhone reports **55 kg**, not ~70. - The laptop source (a commercial carbon-software vendor, [devera.ai](#)) now gives a **median of 215 kg**, not 250.

Draft: Device manufacturing (serves all three profiles)

Evidence (checked directly)

| Device | Source | Life-cycle carbon | Share from making it | Assumed use period |
|----------------|---|--|--|---|
| Phone | Apple, <i>iPhone 17 Product Environmental Report</i> , Sept 2025 (PDF) | 55 kg CO₂e (256 GB), 61 kg (512 GB) | ~76% (53% materials and processes, 23% production electricity). Charging 18%, transport 4%. | 3 years |
| Laptop | Apple, <i>MacBook Air with M4 Product Environmental Report</i> , Mar 2025 (PDF) | 155 kg CO₂e (15-inch, 512 GB) | 71% . Use 25%, transport 3%. | 4 years |
| Laptop (range) | devera.ai benchmark, updated 14 Mar 2026 (page) | Median 215 kg , P10–P90 158–287 kg (10,000 Monte Carlo runs on Ecoinvent 3.9.1 data) | n/a | n/a |
| TV | Samsung, <i>QLED Q60C Product Environmental Report</i> , 28 Dec 2023 (PDF) | 982 kg CO₂e for a 75-inch TV | 70.4% (~690 kg). Use 26.5%. | 7 years. Calculated by the Carbon Trust under PAS 2050. |

What this shows: - For phones, laptops and TVs, making the device accounts for **~70–76% of its lifetime carbon**. That's consistent with the IEA's ~80% for mobile devices (S21). - **Size and configuration matter a lot.** The iPhone varies 55–134 kg across models (Pro Max 2 TB, per search results but not verified). A 75-inch TV is ~1 tonne. A smaller TV would be much less, but **I haven't found a verified figure for a typical 50–55-inch TV**. - **How the calculator could spread manufacturing over use:** divide manufacturing carbon by expected hours of use. For example, ~42 kg for an iPhone 17 over 3 years at 3 hours a day **≈ ~13 g CO₂e per hour of use**.

That's my illustrative arithmetic, not a sourced figure. It depends entirely on the assumed lifetime and daily hours, which would need to be user inputs or clearly stated assumptions.

Limitations and contrary points

- **Manufacturer self-reports.** Apple's life-cycle method is ISO-aligned but not independently published per product. Apple's own report acknowledges "inherent uncertainty in modeling carbon emissions due primarily to data limitations." The Samsung figure was calculated by the Carbon Trust.
- **These show only the best-documented brands and models.** Products from brands that don't publish reports could differ.
- **The use period assumptions differ** (3, 4 and 7 years), so the per-hour spread changes with how long people actually keep devices.
- **Water from manufacturing isn't quantified** in these reports. The calculator's existing phone water figure (~12,800 L) comes from a UK phone-recycling company's blog, which I haven't evaluated yet.
- **I couldn't retrieve Clément, Jacquemotte & Hilty (2020)**, an independent peer-reviewed review of variation in phone and tablet life-cycle studies (*Environmental Impact Assessment Review* 84:106416, [DOI](#)). Its repository link returned a web page instead of the PDF, so I haven't used it.

Assumptions still to check

- **A verified figure for a typical 50–55-inch TV and for a desktop PC and monitor** (Jordan). Dell publishes per-product footprints, but I found only 2010s-era ones.
- **How long employees actually keep devices**, and how many hours a day they use them.
- **Whether to count manufacturing at all** in an activity-based calculator, since the devices are owned anyway. This is the same methodology choice raised in section 5, and yours to make.

Proposed source decisions: - **Apple reports:** use with qualifications. - **Samsung report:** use with qualifications, as a large-TV example. - **devera.ai:** use with qualifications for the laptop range, as a commercial source. - **EPA TV document:** reject as support for "~350 kg."

Question: Should I add this to [research.md](#) as section 9, or first look for a typical-size TV figure and a current desktop PC figure?

25. Learner (RG Sureshkumar)

look for a typical size tv figure and a current desktop pc figure

Agent

[Tool: WebSearch — query: Samsung TV Product Environmental Report 55-inch life cycle carbon emissions kg CO2eq pdf samsung.com sustainability]

Web search results for query: "Samsung TV Product Environmental Report 55-inch life cycle carbon emissions kg CO2eq pdf samsung.com sustainability"

Links: [{"title": "QLED Q60C 12/28/2023 Product Environmental"}]

```
Report", "url": "https://www.samsung.com/global/sustainability/media/pdf/TV_QLED_Environmental_Report_EN.pdf"}, {"title": "Neo QLED QN90C 12/28/2023 Product Environmental Report", "url": "https:// ... [truncated]}
```

[Tool: WebSearch — query: Dell product carbon footprint desktop 2024 2025 OptiPlex OR XPS OR Alienware PCF pdf kg CO2e]

Web search results for query: "Dell product carbon footprint desktop 2024 2025 OptiPlex OR XPS OR Alienware PCF pdf kg CO2e"

Links: [{"title": "Product Carbon Footprint Dell 24 All-in-One EC24250 Report Produced July 2025", "url": "https://www.delltechnologies.com/asset/en-us/products/desktops-and-all-in-ones/technical-support/dell-24-all-in-one-ec24250-desktop-pcf-datasheet.pdf"}, {"title": "Life Cyc ... [truncated]}

[Tool: WebSearch — query: HP OMEN desktop product carbon footprint report kg CO2e]

Web search results for query: "HP OMEN desktop product carbon footprint report kg CO2e"

Links: [{"title": "Product carbon footprint HP 17 Laptop PC Estimated impact 180 – 670† kgCO2e", "url": "https://h20195.www2.hp.com/v2/getpdf.aspx/c07525190.pdf"}, {"title": "Product Carbon Footprint – Results – HP.com", "url": "https://h22235.www2.hp.com/hpinfo/globalcitizenship/environment/productdata/ProductCarbon ... [truncated]}

[Tool: Bash — Extract Dell OptiPlex figures and check HP list for OMEN]

opt.pdf: PDF document, version 1.4, 3 pages
3 5668

[kg CO2] ... he life cycle inventory data for several of the raw and process materials obtained from the background system. Results shown here are subject to change as the model is updated. *Previously stated 257 kg CO2e was a calculation error, this revision reflects the amended figures for this product.

[Manufactur] ... umptions were used to e ... [truncated]

[Tool: Bash — Read OptiPlex life-cycle phase breakdown]

Packaging 2% The selected impact assessment methodology for the life cycle impact assessment (LCIA) is "EF 3.0 Climate Change – total" using the indicator global warming potential 100 (GWP100) measured in CO2-equivalents. Use stage conditions were calculated using Energy Star data. More information about the Energy Star can be found on the Energy Star website. Depending on the end of life allocati ... [truncated]

[Tool: WebSearch — query: 55-inch TV carbon footprint kg CO2e life cycle report LG OLED55 OR Samsung 55 "kg CO2" environmental report manufacturing]

Web search results for query: "55-inch TV carbon footprint kg CO2e life cycle report LG OLED55 OR Samsung 55 "kg CO2" environmental report manufacturing"

Links: [{"title": "LG 2025 OLED Earns Eco-Certifications for Sustainable Design and Reduced

Carbon Footprint | LG Global", "url": "https://www.lg.com/global/newsroom/news/media-entertainment-solution/lg-2025-oled-earns-eco-certifications-for-sustai ... [truncated]

[Tool: Bash — Read Samsung 55-inch signage production share]

```
qmc.pdf: PDF document, version 1.5, 17 pages
[Production] ... UHD Signage QMC Product Environmental Report [1] [2] [3] [2] [3] [4] 2
Product Carbon Footprint Signage QMC Life cycle Sourcing Recycling Production Use Distribution
[6] Signage QM55C Life cycle carbon emissions : 1,691 kg CO2e . [5] 40% 60% 100% Disposal
0.3% Use 76.9 % Production 21.9% Distribution 0.9% 80%20%10% 3 Sourcing Chemical ...
[truncated]
```

I've finished the search. The results are partial: **no consumer 55-inch TV report publishes a figure**, and **no current gaming desktop has a published footprint**. I did find two useful proxies.

TV (typical size) - Samsung, UHD Signage QMC Product Environmental Report ([PDF](#)); Carbon Trust calculation. - The **55-inch QM55C** has a life-cycle total of **1,691 kg CO₂e**, of which **production is 21.9%** and use is 76.9%. - That implies **~370 kg CO₂e to make a 55-inch panel**. That's my arithmetic from Samsung's figures. - **Caveat:** it's a *commercial signage display*, built for long daily running and high brightness, which is why use dominates. It's a proxy for a consumer 55-inch TV, not the same product. - **Coincidentally, it lands near the calculator's current ~350 kg**, but that figure still has no valid source (see the EPA problem above). - **LG and Samsung 2026 consumer TVs** hold Carbon Trust certifications at 55 inches ([LG](#), [Samsung](#)). **Neither announcement publishes the kg figures.**

Desktop PC (current) - Dell, Life Cycle Assessment: OptiPlex Tower Plus 7020, Aug 2024 ([PDF](#)). - **198 kg CO₂e total** (European use), assuming a 4-year life and 47.2 kWh a year. - Made with Dell's ISO 14040-certified modelling tool. The graphics card is 14% of the manufacturing footprint. - **Monitor excluded. - Dell corrected an earlier published 257 kg** as "a calculation error." That's a useful reminder that even official figures move. - **Caveat: this is an office PC** (Core i3, 8 GB RAM, integrated graphics), **not a gaming PC**. A gaming desktop with a large graphics card and power supply would very likely be higher, but by how much isn't evidenced. - **Dell 24 All-in-One** figures (287 ± 82 kg and 313 ± 123 kg, from search results) include a screen but aren't gaming machines either. I haven't opened those documents. - **No footprint reports were found** for Dell Alienware or HP OMEN gaming desktops. The hardware study "Hot Games" (S37) itself had to use Dell's G15 laptop as a proxy because no Alienware assessment exists.

What this means for section 9: - Typical TV: ~370 kg to manufacture, from a commercial-display proxy. Clearly labelled, and probably shown as a range with the 75-inch consumer TV (~690 kg to manufacture). - **Desktop PC:** ~200 kg for an office tower plus a monitor, with an explicit note that **gaming desktops are undocumented** and likely higher.

Question: Should I add section 9 to [research.md](#) now, with these proxies and the problems with the calculator's existing device citations, as sources S39 onward?

26. Learner (RG Sureshkumar)

yes add to research.md

Agent

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

I've added the device manufacturing notes to [research.md](#) as a draft.

- **Audience needs:** a new "9. Device manufacturing (all three profiles)" section. It covers:
 - manufacturing at ~70–76% of lifetime carbon
 - current figures for a phone, laptops and TVs, including the ~370 kg 55-inch proxy
 - the ~200 kg office-desktop figure, with **gaming desktops recorded as undocumented**
 - the illustrative per-hour spreading method
 - the **four problems with the calculator's existing device citations**, listed for correction during implementation
- **Source assessments:** S39 to S44.
 - S39: iPhone 17
 - S40: MacBook Air
 - S41: devera.ai
 - S42: Samsung 75-inch TV
 - S43: Samsung 55-inch signage proxy
 - S44: Dell OptiPlex

Everything is still a proposal for you to confirm. Nothing is marked approved. Whether to count manufacturing at all is recorded as your decision.

That completes the broader digital-life activities (5 to 9). Two cross-cutting topics remain: - **10. Water (Alex).** Water has come up in almost every section with very different accounting: 0.26 mL vs ~10 mL vs 45 mL per prompt. I'd pin down the difference between on-site cooling water and power-plant water, the difference between water withdrawn and water consumed, and whether local water stress (for example Los Angeles vs. where data centers actually are) can be represented honestly. - **11. Trust and communication (Robin).**

Question: Should I go ahead with water?

27. Learner (RG Sureshkumar)

yes go ahead with water

Agent

[Tool: Bash — Read Li et al. Making AI Less Thirsty]

Making AI Less "Thirsty": Uncovering and Addressing the Secret Water Footprint of AI Models
[Li, Pengfei', 'Yang, Jianyi', 'Islam, Mohammad A.', 'Ren, Shaolei'] ['2023/04/06']
Abstract:The growing carbon footprint of artificial intelligence (AI) has been undergoing public scrutiny. Nonetheless, the equally important water (withdrawal and consumption) footprint of AI has largely remained under th ... [truncated]

[Tool: Bash — Look up published version of Li et al. in Crossref]

["Making AI Less 'Thirsty'"] ['Communications of the ACM'] 10.1145/3724499 {'date-parts': [[2025, 6, 27]]} 68 7 54–61
['Thirsty Reservoirs: The Water Footprint Of Hydroelectric Dam Construction'] ['Science Trends'] 10.31988/scitrends.12344 {'date-parts': [[2018, 3, 5]]} None None None
['A Thirsty Land'] None 10.7560/310311 {'date-parts': [[2018]]} None None None

[Tool: WebSearch — query: LBNL 2024 United States Data Center Energy Usage Report water consumption billion gallons direct indirect Shehabi]

Web search results for query: "LBNL 2024 United States Data Center Energy Usage Report water consumption billion gallons direct indirect Shehabi"

Links: [{"title":"Data Center Water Use – MOST Policy Initiative","url":"https://mostpolicyinitiative.org/science-note/data-center-water-use/"}], [{"title":"AI Data Centers and the Water Use Feedback Loop","url":"https://arxiv.org/pdf/2606.21760"}],{"title" ... [truncated]

[Tool: Bash — Extract water figures from LBNL 2024 data center report]

lbnl.pdf: PDF document, version 1.6 (zip deflate encoded)
79 183446
[indirect water] ... t, yet uneven, variability across different regions of the country. Figure 5.10. Water consumption and GHG emission intensity factors of electricity use by county. As noted earlier, U.S. data centers consumed approximately 176 TWh in 2023. The total indirect water footprint of U.S. data centers is nearly 800 b ... [truncated]

[Tool: Bash — Find LBNL direct water consumption figure]

[billion liters] ... 0 as estimated in this report, compared to the “Current Trends” scenario from the 2016 report. Water and Emission Impacts Direct Water Consumption Figure 5.9 shows the growth in direct water usage in data centers. In 2014, data centers consumed 21.2 billion liters of water, with 64% in internal data centers. By 2023, hyperscale and colocation account for 84% of the 66-billion- ... [truncated]

[Tool: Bash — Read abstracts of two data center water geography papers]

The Hidden Water Geography of U.S. Hyperscale Data Centers in the AI Era
[Guidi, Gianluca', 'Dominici, Francesca'] ['2026/06/05']

Abstract:Water use by data centers is routinely reported as a single footprint, but water is consumed through two physically distinct pathways: at the site for cooling and in the power system that generates electricity. We mapped both pathways for 472 U.S. hyperscal ... [truncated]

[Tool: WebSearch — query: USGS domestic water use per capita gallons per day United States estimate]

Web search results for query: "USGS domestic water use per capita gallons per day United States estimate"

Links: [{"title":"How much water is used by people in the United States? | U.S. Geological Survey","url":"https://www.usgs.gov/faqs/how-much-water-used-people-united-states"}, {"title":"Per capita water use: How much water do you use at home? USGS Water Science School","url":"https://water.usg ... [truncated]

[Tool: Bash — Verify USGS per-capita figure and Li et al. withdrawal definition]

LEN 555

2.1 Water Withdrawal vs. Water Consumption There are two related but different concepts – water withdrawal and water consumption, both of which are important for understanding the impacts on water stress and availability [12 , 13] . • \bullet Water withdrawal: It refers to freshwater taken from the ground or surface water sources, either temporarily or permanently, and then used for agri ... [truncated]

[Tool: WebFetch — https://www.usgs.gov/faqs/how-much-water-used-people-united-states]

I've reviewed the provided web page content, but it does not contain specific per-capita domestic water use figures in gallons per day or the year of such an estimate.

The page states that in 2015, total U.S. water use was "322 billion gallons per day (Bgal/day)" with three largest categories being irrigation, thermoelectric power, and public supply. However, no breakdown for per-capita domestic ... [truncated]

[Tool: Bash — Verify USGS 2015 domestic per-capita water use]

usgs.pdf: PDF document, version 1.6 (zip deflate encoded)

16

Public Supply and Domestic Water Use in the United States, 2015 Open-File Report 2017-1131 U.S. Department of the Interior U.S. Geological Survey Cover: Photograph of Montebello Filtration Plant, Baltimore, Maryland, 2015. (Photograph by Cheryl Dieter, U.S. Geological Survey) Public Supply and Domestic Water Use in the United States, 201 ... [truncated]

[Tool: Bash — Find per-capita GPCD statements in USGS report]

[GPCD] ... largest users, with each state delivering about 950 Mgal/d to domestic users from public-water suppliers (Dieter and others, 2017). Per capita water use for domestic deliveries ranged from 35 gallons per capita per day (GPCD) in Connecticut to 186 GPCD in Idaho (Dieter and others, 2017). The national average per capita water use for domestic deliveries was 83 GPCD (Dieter and others, 20 ... [truncated]

Water notes are drafted below. The research explains why the water numbers in earlier sections disagree so much, and it turned up **a consistency problem in the calculator itself**.

Draft: Water (serves Alex; relevant to all)

Evidence (checked directly)

- 1. What counts as "AI's water"** (Li, Yang, Islam & Ren, "Making AI Less 'Thirsty'," *Communications of the ACM* 68(7):54–61, 2025, [DOI](#); [arXiv](#)): - **Scope 1**: on-site water evaporated to cool the data center. - **Scope 2**: off-site water used by power plants to generate its electricity. - **Scope 3**: water used in manufacturing the chips and servers. - **Withdrawal vs. consumption**: - *Withdrawal* is water taken from a source, much of it returned. - *Consumption* is water evaporated or otherwise removed, which affects downstream supply. - AI figures are usually *consumption*.
- 2. In the US, power-plant water far outweighs cooling water**: - Shehabi et al., *2024 United States Data Center Energy Usage Report*, Berkeley Lab, Dec 2024 ([PDF](#)): - US data centers **directly consumed ~66 billion liters in 2023** (84% at hyperscale and colocation sites). - The **indirect footprint through electricity was ~800 billion liters**, about **12x the direct figure**. - Indirect intensity: **4.52 L/kWh** for data centers vs. **4.35 L/kWh** US average. - Average on-site water use per kWh was **~0.36 L/kWh** in 2023. - Guidi & Dominici, "The Hidden Water Geography of U.S. Hyperscale Data Centers in the AI Era," arXiv, June 2026 ([arXiv](#)): - Across 472 hyperscale sites, **electricity-related water is about three-quarters** of the ~300 GL per year operational total (range 205–451).
- 3. Where the water is used matters as much as how much**: - Guidi & Dominici: **cooling burdens cluster in water-stressed western and south-central basins**, while **electricity-related burdens cluster in a few eastern, fossil-heavy grid regions**. Just 3 of 24 grid operators account for 59% of electricity-related water. - Akinade, Amanambu, Frame & Ren, arXiv, June 2026 ([arXiv](#)): community-level burden across ten US sites spans **0.2% to 134% of local utility capacity**. I read the abstract only.
- 4. Household baseline**: US average domestic use was **82 gallons (~310 L) per person per day** in 2015 (Dieter & Maupin, USGS Open-File Report 2017–1131, [PDF](#)).

Why the per-prompt water figures in earlier sections disagree

| Figure | What it counts |
|---|---|
| 0.26 mL (Google, S14) | Scope 1 only: on-site cooling |
| ~1–22 mL for a 300-word reply (EcoLogits, calculator) | Scopes 1 + 2: cooling plus power plants (Li method) |
| 45 mL (Mistral, S16) | Full life cycle, including manufacturing |

Since power-plant water is ~12x cooling water in the US, **leaving out scope 2 shrinks the figure by roughly an order of magnitude**. Neither number is "wrong." They answer different questions.

Limitations and contrary points

- **On-site intensity estimates disagree**. Li et al. cite **1–9 L/kWh evaporated** (1 L for Google's global average, 9 for an Arizona data center in summer). Berkeley Lab's modelled US average is **~0.36 L/kWh**,

and it notes some hyperscalers report 0.1–0.3 for certain systems. This gap is unresolved.

- **Power-plant water factors are averages.** They ignore renewable power contracts (Berkeley Lab says so explicitly), and they depend on the grid, which varies by county.
- **Nobody can know which data center serves a given request.** So the calculator can't honestly place Alex's AI water in Los Angeles, or anywhere else.
- **Guidi & Dominici and Akinade et al. are preprints.** The latter includes Shaolei Ren, a co-author of Li et al. and a prominent critic of Google's figures (S17).

A consistency problem in the calculator (for your judgement)

- **The calculator mixes accounting methods.** Its AI water (EcoLogits) is *consumption* for cooling and power. Its everyday comparisons use **Water Footprint Network** figures: coffee ~140 L, a burger ~1,700 L, a phone ~12,800 L ([index.html](#)). Those are whole-supply-chain footprints.
- **Water Footprint Network figures typically include rainfall absorbed by crops ("green water")** as well as irrigation and pollution-dilution water. I haven't verified that definition in this session. If it holds, the comparisons set AI's narrow consumption figure against very broad food footprints.
- **The household baseline (82 gal/day) is delivered water**, which is mostly returned via sewers, **not consumption**.
- **Like-for-like comparisons would need a consistent definition.** How to handle that is a methodology decision for you.

Assumptions still to check

- **The Water Footprint Network's green/blue/grey definitions**, checked directly.
- **Whether water stress can be shown honestly without knowing data-center locations.** One option is to explain the geography qualitatively rather than as a number. That's your call.

Proposed source decisions: - Li et al. (CACM 2025): use. - Berkeley Lab 2024: use. - Guidi & Dominici: use with qualifications. - Akinade et al.: background. - USGS: use with qualifications, noting it measures delivery, not consumption.

Question: Should I add this to [research.md](#) as section 10, or first verify the Water Footprint Network definitions behind the calculator's comparisons?

28. Learner (RG Sureshkumar)

verify the water footprint network definitions

Agent

[Tool: Bash — Read Water Footprint Network definitions and calculator's cited page]

```
### https://www.waterfootprint.org/water-footprint-2/what-is-a-water-footprint/
```

```
HTTP 200
```

```
</html><!-- WP Fastest Cache file was created in 1.098 seconds, on September 1, 2026 @ 10:39
```

am -->LEN 7478

[Green water] ... humanity. This also includes the direct water footprint, which is the water used directly by the individual(s) and the indirect water footprint – the summation of the water footprints ... [truncated]

The definitions are confirmed: **the calculator's food and product water figures mostly count water that isn't comparable to AI's water.**

The Water Footprint Network's definitions ([page](#)): - **Green water**: "water from precipitation that is stored in the root zone of the soil and evaporated, transpired or incorporated by plants." In other words, rainfall used by crops. - **Blue water**: water "sourced from surface or groundwater resources" that is evaporated, built into a product, or returned elsewhere or later. **This is the category comparable to AI's consumption figures.** - **Grey water**: "the amount of fresh water required to assimilate pollutants to meet specific water quality standards." A notional volume for diluting pollution, not water actually used up.

What this means for the calculator's comparisons:

Calculator item	Its figure	What's inside it (verified)	Blue water only
Beef burger	~1,700 L (at 15,400 L/kg)	Water Footprint Network: beef $\approx$ 15,000 L/kg, 93% green, 4% blue, 3% grey	~70 L (4%, my arithmetic)
Smartphone	~12,800 L (UK recycling-company blog)	GRACE Communications' Water Footprint Calculator (page): ~3,190 gallons (~12,100 L) per phone, of which grey water "makes up the largest portion"	Not published
Cup of coffee	~140 L (World Economic Forum article)	Not checked. Coffee footprints are usually mostly green water.	Not checked

The consequence: - **The current comparisons overstate everyday items relative to AI.** For example, the calculator sets ~10 mL of blue water for a GPT-5.5 reply against a burger's 1,700 L. That figure is almost entirely rainfall on feed crops. - **Comparing blue with blue shrinks the gap a lot.** The burger falls from ~1,700 L to ~70 L. It's still thousands of times a single reply, but by a factor of ~25 less than the calculator currently shows. - **This affects the brief's neutrality requirement.** Inconsistent accounting can make AI look either trivial or significant, depending on which figures are paired. It's worth recording as a correction for implementation.

One more note: the smartphone figure's grey water is a notional volume for diluting pollution, not water used up. Presenting it alongside consumption figures without explanation would mislead in the other direction.

Proposed decisions: - **Water Footprint Network definitions page:** use, as the authority for definitions and the beef breakdown. - **GRACE Communications' calculator page:** use with qualifications. - **Flag for implementation:** use blue water only (or clearly labelled categories) for like-for-like comparisons. **How to fix this is your decision.**

Question: Should I add the water section to [research.md](#) as section 10, including these definitions and the comparison problem, as sources S45 onward?

yes add it to research.md

Agent

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.

I've added the water notes to [research.md](#) as a draft.

- **Audience needs:** a new "10. Water (Alex; relevant to all profiles)" section. It covers:
 - the three scopes of AI water, and withdrawal vs. consumption
 - power-plant water at ~12x cooling water in the US
 - where the water is used
 - why the earlier per-prompt water figures differ
- the **calculator's inconsistent comparisons:** AI's blue water against mostly-green burger water and mostly-grey phone water, recorded for correction during implementation, with the fix left to you
- **Source assessments:** S45 to S51.
 - S45: Li et al.
 - S46: Berkeley Lab
 - S47: Guidi & Dominici
 - S48: Akinade et al.
 - S49: USGS
 - S50: Water Footprint Network
 - S51: GRACE Communications

Everything is still a proposal for you to confirm. Nothing is marked approved.

The last topic is **11. Trust and communication (Robin)**, who is skeptical of the company's motives and wants plain language, visible sources, and honest uncertainty. This is different from the footprint research: I'd look for evidence on: - how to communicate uncertainty and ranges to non-experts so they're understood and trusted, for example research on verbal vs. numerical uncertainty and on presenting ranges - how disclosing funding or sponsorship affects trust - whether employer-provided environmental tools are viewed with suspicion

Question: Should I go ahead with trust and communication?

30. Learner (RG Sureshkumar)

yes go ahead with trust and communication

Agent

[Tool: Bash — Fetch metadata and abstracts for van der Bles papers]

```
['The effects of communicating uncertainty on public trust in facts and numbers'] ['van der Bles', 'van der Linden', 'Freeman', 'Spiegelhalter'] ['Proceedings of the National Academy of Sciences'] 117 14 7672-7683 {'date-parts': [[2020, 3, 23]]}
```

Uncertainty is inherent to our knowledge about the state of the world yet often not communicated alongside scientific facts and numbe ... [truncated]

[Tool: WebSearch — query: meta-analysis effects of uncertainty communication on trust credibility Gustafson Rice 2020 Public Understanding of Science]

Web search results for query: "meta-analysis effects of uncertainty communication on trust credibility Gustafson Rice 2020 Public Understanding of Science"

Links: [{"title": "A review of the effects of uncertainty in public science communication", "url": "https://www.researchgate.net/publication/343035909_A_review_of_the_effects_of_uncertainty_in_public_science_communication"}, {"title": "The impact o ... [truncated]

[Tool: WebSearch — query: experiment disclosure of industry funding effect on trust in science communication credibility study]

Web search results for query: "experiment disclosure of industry funding effect on trust in science communication credibility study"

Links: [{"title": "Science communication gets personal: Ambivalent effects of self-disclosure in science communication on trust in science - PubMed", "url": "https://pubmed.ncbi.nlm.nih.gov/37589716/"}, {"title": "From Participation to Trust? Understanding Trust Dynamics ... [truncated]

[Tool: Bash — Fetch abstracts for uncertainty review, meta-analysis, and disclosure study]

```
['A review of the effects of uncertainty in public science communication'] ['Gustafson', 'Rice'] ['Public Understanding of Science'] 29 6 614-633 {'date-parts': [[2020, 7, 17]]}
```

Uncertainty is inherent to science and science communication. However, the evidence appears mixed regarding whether portraying uncertainty in science communication has positive or negative effects. We review a diverse ran ... [truncated]

[Tool: WebSearch — query: employee skepticism corporate sustainability communication greenwashing perception employees trust study peer-reviewed]

Web search results for query: "employee skepticism corporate sustainability communication greenwashing perception employees trust study peer-reviewed"

Links: [{"title": "Perceived greenwashing and employee green behavior: The roles of green

organizational identity and self-serving leadership – Ma – 2025 – Business Ethics, the Environment & Responsibility – Wiley Online Library", "url": "https://onli ... [truncated]

[Tool: Bash — Fetch abstracts for employee greenwashing studies]

['Perceived greenwashing and employee green behavior: The roles of green organizational identity and self-serving leadership'] ['Ma', 'Zhao', 'Chen', 'Hu', 'Qu'] ['Business Ethics, the Environment & Responsibility'] 34 4 1475–1486 {'date-parts': [[2024, 7, 28]]}
Abstract Greenwashing has become a common way in which many enterprises address expectations related to corporate social responsibili ... [truncated]

[Tool: WebSearch — query: Cain Loewenstein Moore 2005 "The Dirt on Coming Clean" abstract disclosure advisors estimators experiment]

Web search results for query: "Cain Loewenstein Moore 2005 "The Dirt on Coming Clean" abstract disclosure advisors estimators experiment"

Links: [{"title": "(PDF) DAYLIAN M. CAIN GEORGE LOEWENSTEIN DON A. MOORE", "url": "https://www.academia.edu/2839968/DAYLIAN_M_CAIN_GEORGE_LOEWENSTEIN_DON_A_MOORE?hb-g-sw=2079004"}, {"title": "(PDF) The dirt on coming clean: Perverse effects of disclosing conflicts o ... [truncated]

[Tool: Bash — Verify Cain et al. abstract on IDEAS]

LEN 11781
[Abstract] ... n: Perverse Effects of Disclosing Conflicts of Interest Author & abstract Download 6 References 147 Citations Most related Related works & more Corrections Author Listed: Daylian M. Cain George Loewenstein Don A. Moore Registered: George Loewenstein Abstract Conflicts of interest can lead experts to give biased and corrupt advice. Although disclosure is often proposed as a ... [truncated]

[Tool: Bash — Fetch Budescu et al. 2009 IPCC verbal uncertainty abstract]

['Improving Communication of Uncertainty in the Reports of the Intergovernmental Panel on Climate Change'] ['Budescu', 'Broomell', 'Por'] ['Psychological Science'] 20 3 299–308 {'date-parts': [[2009, 3]]}
The Intergovernmental Panel on Climate Change (IPCC) assesses information relevant to the understanding of climate change and explores options for adaptation and mitigation. The IPCC reports comm ... [truncated]

I've checked the trust and communication evidence. Unlike the footprint sections, these are peer-reviewed studies about people. Draft below.

Draft: Trust and communication (serves Robin)

Evidence (checked directly)

1. Showing uncertainty as numbers or ranges doesn't meaningfully damage trust. - van der Bles, van der Linden, Freeman & Spiegelhalter, *PNAS* 117(14):7672–7683, 2020 ([DOI](#)). - 5 experiments, including a

preregistered national replication and a field experiment on BBC News (n = 5,780). - Communicating uncertainty produced "only a small decrease in trust in numbers and trustworthiness of the source, and **mostly for verbal uncertainty**." - Gustafson & Rice, *Public Understanding of Science* 29(6):614–633, 2020 ([DOI](#)). - A review of 48 experiments. **Quantified error ranges and probabilities had "only positive or null effects."** - Negative effects came mostly from **uncertainty framed as scientists disagreeing** ("consensus uncertainty"). That framing never had positive effects. - Prior beliefs and worldviews moderated the effects. - Dries, Schneider, Oeberst, McDowell, Egharevba & Rebitschek, *Royal Society Open Science* 13(4), 2026 ([DOI](#)). - A meta-analysis of 28 studies. Uncertainty communication "neither systematically undermined nor enhanced trust" in the source. - It notes most studies used fictitious contexts, not real communicators.

2. Vague words for uncertainty are widely misread. - Budescu, Broomell & Por, *Psychological Science* 20(3):299–308, 2009 ([DOI](#)). - When people put numbers on the IPCC's words ("likely," "very likely"), their answers deviated significantly from the IPCC's own definitions, **even when they had the definitions in front of them.**

3. Disclosing a conflict of interest doesn't automatically fix trust. - Cain, Loewenstein & Moore, "The Dirt on Coming Clean," *Journal of Legal Studies* 34(1):1–25, 2005 ([DOI](#); abstract via [IDEAS](#)). - People "do not discount advice from biased advisors as much as they should" even after disclosure, and **disclosure can make advisors exaggerate more.**

4. Employees distrust sustainability messaging they see as greenwashing. - Mu & Lee, *Sustainability* 15(22):15693, 2023 ([DOI](#)): among 304 employees, perceived greenwashing was **negatively related to trust** and, through trust, to loyalty. - Ma et al., *Business Ethics, the Environment & Responsibility* 34(4):1475–1486, 2024 ([DOI](#)): among 232 employees, perceived greenwashing was **negatively related to employees' own green behavior.**

What this suggests for Robin's stated needs (for your judgement)

- **Show ranges as numbers, not vague words.** This is the best-supported finding. Ranges carry little trust cost, while vague words are both less trusted and misread. This fits the earlier sections, where most figures are genuinely ranges.
- **Present disagreement carefully.** Sources that really do conflict (Google vs. its critics, TikTok vs. Greenly, the CogVideoX figures) are "consensus uncertainty," the kind most linked to lost trust. The evidence doesn't say to hide these disagreements. The brief requires honesty. But explaining *why* sources differ (different boundaries and methods), as in sections 4 and 10, may be better than simply showing competing numbers.
- **Disclose funding, but don't treat it as enough on its own.** Examples include Netflix funding the Carbon Trust study (S25), Publicis funding EcoLogits' video work (S6), and the company's own role. Disclosure is ethically required, but Cain et al. suggest it doesn't by itself make an audience trust appropriately. Visible sources and methods probably matter more.
- **An employer-provided tool starts with a credibility risk.** If employees see it as promotional, trust and engagement fall. That supports the brief's rule that the calculator must argue neither that AI is harmless nor that it is harmful.

Limitations

- **Most uncertainty experiments use fictitious scenarios and general audiences**, not employees using a company's own tool (Dries et al.).
- **The greenwashing studies are surveys of employees in China**, measuring associations, not causes. They may not transfer to a US creative-media company.
- **Cain et al. study advisors paid to bias estimates**, which is a different situation from a company publishing a calculator.
- **No study was found on employer-provided environmental calculators specifically.**
- **Robin is a fictional profile.** These findings describe tendencies, not what any particular skeptical employee will think.

Proposed source decisions: - van der Bles 2020, Gustafson & Rice 2020, Dries et al. 2026, Budescu et al. 2009: use. - Cain et al. 2005: use with qualifications. - Mu & Lee 2023, Ma et al. 2024: use with qualifications.

Question: Should I add this to [research.md](#) as section 11, as sources S52 onward? That would complete the Audience needs research, and the next step would be drafting **Possible features**.

31. Learner (RG Sureshkumar)

yes add it to research.md and we can start drafting the possible features

Agent

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

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[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/research.md has been updated successfully.

[Tool: Bash — Get authors' given names from Crossref]

```
10.1098/rsos.251896 [('Charlotte', 'Dries'), ('Claudia R.', 'Schneider'), ('Aileen', 'Oeberst'), ('Michelle', 'McDowell'), ('Gabriel', 'Egharevba'), ('Felix G.', 'Rebitschek')]
10.3390/su152215693 [('Honglei', 'Mu'), ('Youngchan', 'Lee')]
10.1111/beer.12723 [('Yufei', 'Ma'), ('Shuming', 'Zhao'), ('Jiaxi', 'Chen'), ('Chenhong', 'Hu'), ('Jiaojiao', 'Qu')]
10.1111/j.1467-9280.2009.02284.x [('David V. ... [truncated]
```

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/research.md has been updated successfully.

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/research.md has been updated successfully.

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/research.md has been updated successfully.

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/research.md has been updated successfully.

[Tool: Bash — Verify author given names for two more citations]

```
10.1073/pnas.1913678117 [('Anne Marthe', 'van der Bles'), ('Sander', 'van der Linden'), ('Alexandra L. J.', 'Freeman'), ('David J.', 'Spiegelhalter')]
10.1177/0963662520942122 [('Abel', 'Gustafson'), ('Ronald E.', 'Rice')]
```

I've added section 11 to [research.md](#) as sources S52 to S58. That completes the Audience needs research.

A correction I made along the way: when first writing S54 and S58, I filled in some author initials that I hadn't verified. For example, I wrote "Schneider, R." where the correct name is Claudia R. Schneider. I've now checked every author name for those entries against Crossref and fixed them.

Draft: Possible features

Here are 12 candidates, kept brief as research.md asks. Each is meant to change what the calculation does, since the brief says styling and wording changes don't count. The **P / D / X** tags show which brief requirement each could satisfy: - **P**: professional AI use - **D**: broader digital life - **X**: cross-cutting, a candidate for the "strongest remaining need"

#	Idea	Helps someone learn or do	Serves	Evidence or implementation challenge
1	Image generation logging (P)	Log images by model type and settings (resolution, quality/steps), and see them in the day's total	Alex	Only open-model figures exist (0.09–11.5 Wh per image). No commercial-model data. Water would have to be derived (§1).
2	Video generation logging (P)	Log clips by tool, length and resolution, using EcoLogits' video method	Alex	Commercial figures are <i>estimates</i> (Veo ~20–43 Wh, Sora 2 Pro ~315–534 Wh per 8-second clip).

#	Idea	Helps someone learn or do	Serves	Evidence or implementation challenge
				The EcoLogits calculator's outputs still need checking against the paper (§2).
3	Attempts per final output (P)	Record how many drafts or variations go into one finished image, clip or task	Alex, Jordan	Variation is well evidenced (up to 30x for coding tasks). The ad example (70,000 videos) is unverified. Could be folded into 1 and 2.
4	Agent session and project estimator (P)	Estimate coding-agent sessions from their length and makeup (fresh input, cached input, output tokens) and sum them across a project	Jordan	Fixes the current output-only method (§3). Energy for cached tokens is unknown, so it would need an explicit wide range.
5	Digital day builder (D)	Enter hours of video calls, streaming, social media and gaming, by device (phone, laptop, TV, console, PC)	All three	The device dominates in every source (TV $\approx$ 90x a phone). Social media and music have weak evidence (§5–8).
6	Device manufacturing spread over use (D)	Enter the devices you own and how long you keep them, and see manufacturing emissions per day of use	All three	Your methodology decision on whether to count manufacturing at all. Gaming PCs are undocumented. TV figure is a proxy (§9).
7	AI in context of your digital day (D)	See AI as a share of your <i>total</i> digital footprint, calculated from 1/4 + 5 (+6)	All three	Depends on 5. Must avoid framing that implies AI is trivial or dominant.
8	Water boundary selector (X)	Switch between cooling-only, cooling plus power plants, and full life cycle, and see why per-prompt water varies 100x	Alex, Robin	Well evidenced (§4, §10). Needs consistent definitions for every item.
9	Like-for-like comparisons (X)	Compare against everyday items using matching definitions (for example blue water vs. blue water), fixing the burger and phone mismatch	Alex, Robin	The fix is your choice. Coffee's composition hasn't been checked (§10).
10	Ranges carried through every result (X)	Every total shows low–high numbers, not a single figure, carried through all calculations	Robin, Jordan	The strongest trust evidence supports numeric ranges (§11). Needs a range for every input.
11	"Why sources differ" and disclosure panel (X)	Beside each figure, see its boundary, method, funding and any disagreement between sources	Robin	The evidence favours explaining disagreement over showing competing numbers. Partly a presentation feature, so it would need to be tied to the calculation, for example by selecting which source feeds the total.
12	Home grid region selector (D/X)	Choose a region (for example the LA, Austin or Chicago grid) for the carbon from your <i>home devices</i>	All three	Grid factors matter a lot. Not yet researched (US regional grid data not checked). It can't honestly localise the data centers behind AI requests.

Early observations (not choices): - **Professional AI:** 1–4 are the natural candidates. 3 might merge into 1 and 2. - **Digital life:** 5 and 6 are the strongest. 7 depends on 5. - **Strongest remaining need:** 8–11 all address it. **Trust and consistent accounting (Robin, plus the calculator's own inconsistencies)** looks like the biggest gap the research found, and 10 is the best evidenced. - **Needs more research first:** 12.

Question: Before I write these into [research.md](#), do you want to add, merge or drop any ideas? For example, merge 3 into 1 and 2, or drop 12 unless we research grid data.

32. Learner (RG Sureshkumar)

no the ideas look good we can add them to research.md

Agent

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

```
The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.
```

I've added all 12 ideas to the **Possible features** section of [research.md](#). Each has its tag, what it would help someone learn or do, who it serves, and its evidence or implementation challenges, plus the early observations. **The section is marked as a draft, and no idea is marked as selected.**

What's left in this stage: 1. **Selected features:** five features, why each was chosen, how the set serves all three profiles, and a few serious alternatives with reasons for rejecting them. **The choice is yours.** I can draft the rationale once you've picked. 2. **Your review of the whole document**, including confirming or changing the proposed source decisions (S1–S58), and your **explicit approval** of the selected features. 3. Then, after your approval, the **research transcript**.

The brief's constraints on the five: - At least **2 professional AI** (from 1–4) - At least **2 broader digital life** (from 5–7, 12) - The **5th** must meet the strongest remaining need (8–11 are the candidates) - Together they must serve **Alex, Jordan and Robin**

Question: Which five do you want to select? If you'd rather first see a short side-by-side comparison of the strongest options, or have idea 12's grid data researched, tell me that instead.

33. Learner (RG Sureshkumar)

show me a short side by side comparison of the strongest options

Agent

Here's a side-by-side comparison of the strongest options in each slot. Ratings are my assessments from the evidence in [research.md](#). **These are not recommendations; the choice is yours.**

Professional AI (need at least 2)

	1. Image logging	2. Video logging	4. Agent/project estimator
Serves	Alex	Alex	Jordan
Evidence strength	Moderate: open models only, 46x+ spread	Moderate to low: commercial figures are <i>estimates</i>	Low to moderate: input-token range ~1–4 Wh per 10k, cached tokens unknown
Fills a real gap?	Yes. The calculator has no images today.	Yes. No video today, and video can be far above text.	Yes. It fixes a flaw in the current output-only method.
Build effort	Low to medium (new data table)	Medium (needs EcoLogits video figures extracted and checked)	Medium to high (new inputs plus a range model)
Main risk	Shows open-model figures for commercial tools	Precise-looking numbers for guesses	Wide ranges may feel unhelpful

Idea 3 (attempts per output) adds little on its own. It works best as an input inside 1, 2 or 4.

Broader digital life (need at least 2)

	5. Digital day builder	6. Device manufacturing	7. AI as a share of digital day
Serves	All three	All three	All three
Evidence strength	Strong for streaming, consoles, TVs. Weak for social media and music.	Moderate: manufacturer reports; TV proxy; no gaming PC data	Inherits from 5 (and 6)
Fills a real gap?	Yes. No digital activities can be entered today.	Yes. Manufacturing is ~70–76% of device carbon.	Partly. It's a view built on 5.
Build effort	Medium	Medium	Low once 5 exists
Main risk	Showing weak categories as precise	Needs your decision on whether to count manufacturing at all	Could be seen as a framing device rather than a feature

Strongest remaining need (the 5th)

	8. Water boundary selector	9. Like-for-like comparisons	10. Ranges everywhere	11. "Why sources differ" panel
Serves	Alex, Robin	Alex, Robin	Robin, Jordan (and all)	Robin
Evidence strength	Strong (§4, §10)	Strong (definitions verified)	Strongest (three peer-reviewed syntheses, §11)	Moderate (explaining disagreement is favoured, but less directly tested)
Fixes a known flaw?	Partly	Yes. Burger and phone comparisons are currently mismatched.	Partly (the current figures show mostly single values)	No
Counts toward the calculation?	Yes	Yes	Yes	Only if it changes which source feeds the total
Build effort	Medium	Low to medium	Medium to high (every input needs a range)	Medium
Main risk	Complexity for casual users	Requires changing existing comparisons	Effort across the whole app	May not count as a "meaningful feature"

Points that may help you decide

- **Coverage:** Alex is served by 1 and 2. Jordan by 4. Robin mainly by 8–11. Everyone by 5 and 6. A set with no feature from 8–11 would lean on 5 and 6 for Robin.
- **Merging is possible.** For example, 3 folded into 1, 2 and 4, or 8 and 9 combined as one "consistent water accounting" feature. Whether a merged feature counts as one "meaningful feature" under the brief is your judgement.
- **Open decisions attach to some options.** 6 depends on whether to count manufacturing. 9 depends on how to fix the comparisons. 5 depends on how to handle weak categories.

Question: Which five do you want to select, and do you want to combine any?

34. Learner (RG Sureshkumar)

I will move forward with image logging, video logging, digital day builder, device manufacturing, and like for like comparisons

Agent

You've chosen: **1. Image logging, 2. Video logging, 5. Digital day builder, 6. Device manufacturing, and 9. Like-for-like comparisons.** Here's how the set measures against the brief.

Brief requirements: - **At least 2 professional AI features:** met by 1 and 2. - **At least 2 broader digital life features:** met by 5 and 6. - **5th meets the strongest remaining need:** met by 9, which fixes the mismatched comparisons that §10 found (a burger that's ~93% rainfall water, a phone that's mostly notional dilution water). - **Serves all three profiles:** met, but coverage is uneven: - **Alex:** strongly served (images, video, streaming and social media by device, water comparisons). - **Robin:** served through 5 (video meetings, streaming, social media), 6 and 9. Consistent comparisons support Robin's need for honest numbers. But Robin's stated need for **visible uncertainty** isn't the main focus of any selected feature (that was idea 10). - **Jordan:** served through 5 (video calls, gaming) and 6 (desktop PC). But Jordan's **main stated need is coding agents, long sessions, ranges and project totals, and no selected feature addresses it.** The flaw in the current output-only agent figure (§3) would also stay as it is. Jordan's gaming-PC data is also the weakest in 5 and 6 (no current independent measurement; no gaming-desktop manufacturing report).

This is your decision, and the set does meet the brief. I'm flagging the Jordan gap because the Selected features section has to explain "how the set serves all three reference profiles." A reviewer is likely to ask about it, and it's worth deciding now how to answer. Some options: - Keep the set, and name idea 4 (and 10) explicitly as serious alternatives, with your reason for rejecting them. - Keep the set, and record the output-only agent figure as a known limitation to disclose in the calculator. - Change the set.

Your open decisions attached to this set (these can be settled in the specification stage, but belong to you): - **6:** whether and how manufacturing counts (and gaming desktops are undocumented). - **9:** how the comparisons will be corrected (for example blue water only, or labelled categories). - **5:** how to present weak categories (social media, music).

Question: How do you want to handle the Jordan gap? Keep the set and document it (and if so, what's your reason for not choosing idea 4?), or change the selection?

35. Learner (RG Sureshkumar)

actually lets drop the video logging and do agent/project estimator instead

Agent

Updated set: **1. Image logging, 4. Agent/project estimator, 5. Digital day builder, 6. Device manufacturing, and 9. Like-for-like comparisons.**

Brief check: - **Professional AI:** met by 1 and 4. - **Broader digital life:** met by 5 and 6. - **5th, strongest remaining need:** met by 9. - **All three profiles:** met, and more evenly than before. - **Jordan:** now has a feature for the main stated need (4: sessions, ranges, project totals), plus 5 (video calls, gaming) and 6 (desktop PC). It also replaces the flawed output-only agent figure. - **Alex:** 1 (images), 5 (streaming and social media by device), 6 and 9 (water). - **Robin:** 5 (video meetings, streaming, social media), 6, and 9 (consistent, honest comparisons).

The trade-off to acknowledge: with video logging dropped, **Alex's video-generation use isn't represented**, although the profile names it explicitly. The research supports a reason you could give: video estimates for commercial tools are inferred, not measured (S5), sources conflict (S2 vs S4), and the EcoLogits outputs haven't been checked. The reason is your call, though, and it will need to appear in the **rejected alternatives**.

Next step: drafting the Selected features section. It needs, for each of the five, why it was selected, how the set serves all three profiles, and a few serious alternatives with reasons for rejecting them. I'd draft it from the research, clearly marked as a draft for your review. **Proposed rejected alternatives and reasons, for you to confirm or rewrite:**

Alternative	Draft reason for rejecting (from the research)
2. Video logging	Commercial-tool figures are estimates from assumed hardware, and sources conflict. Too uncertain to present as a logged value right now.
10. Ranges everywhere	Best-supported by the trust evidence, but it would need a range for every input across the whole calculator. Too large for one feature. Parts of it (ranges in 4) are included.
8. Water boundary selector	Overlaps with 9. Consistent definitions (9) are a prerequisite for a meaningful selector.
11. "Why sources differ" panel	Mostly presentation. It would count only if tied to the calculation.
7. AI share of digital day	Depends on 5 and is closer to a view than a separate feature.
3. Attempts per output	Adds little on its own. Can be an input inside 1 and 4.
12. Grid region selector	Not researched yet (US regional grid data unchecked).

Question: Do these reasons reflect your thinking? Confirm them, or tell me what to change (especially for video logging), and I'll draft the Selected features section.

36. Learner (RG Sureshkumar)

yes this works, go ahead and draft the selected features section

Agent

[Tool: Bash — Locate section headings in research.md]

```
7:## Instructions for the user
11:## Instructions for the agent
17:## Reference employee profiles
25:## Audience needs
314:## Possible features
339:## Source assessments
813:## Selected features
817:User approval: Review the completed research directly. Confirm that sources exist and
support the claims the project will use, correct the document as needed, and explicitly
approve the selected featur ... [truncated]
```

[Tool: Read — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

```
812
813 ## Selected features
814
815 List the five selected features. Briefly explain why each was selected and how the set
serves all three reference profiles. Name a few serious alternatives and explain why they were
rejected.
816
817 User approval: Review the completed research directly. Confirm that sources exist and
support the claims the project will use, correct the document as needed, a ... [truncated]
```

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

```
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Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.
```

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/research.md]

```
The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI
Explorations/Unit 1 Brief - AI Footprint Calculator/research.md has been updated successfully.
```

I've drafted the **Selected features** section in [research.md](#), marked as a draft of your selection and **not approved**. It contains:

- **The five features, each with why it was selected:** 1. Image generation logging 2. Agent session and project estimator 3. Digital day builder 4. Device manufacturing spread over use 5. Like-for-like comparisons
- **How the set serves each profile**, plus an explicit check against the brief's requirements.
- **Three known gaps**, stated openly:
 - Alex's video generation isn't covered.
 - Robin's uncertainty need is only partly covered.
 - Jordan's gaming-PC data is thin.
- **Seven rejected alternatives**, each with the reason you confirmed.

What's needed from you before the research can be approved: 1. **Review the whole document directly.**

It's long: 11 audience-need sections, 12 possible features, and 58 source assessments. The instructions ask you to confirm that the sources exist and support the claims, and to correct anything you disagree with. 2. **Confirm or change my proposed source decisions**, especially the two I left for you: - **S15 (Altman)**: use as an unsupported company claim, or reject? - **S29 (Greenspector 2023)**: use with strong qualifications, or reject? 3. **Update or remove the "Status: draft" markers** once you're satisfied. I won't mark anything approved. 4. **Explicitly approve the selected features.**

Methodology decisions you've deferred to the specification stage don't block approval, but they're recorded in the document: - whether and how manufacturing counts - how the comparisons will be corrected - how weak categories are presented - whether to count the user's own device

Question: Do you want to review [research.md](#) now and come back with corrections or your approval, or would you like me to walk you through any section first?

37. Learner (RG Sureshkumar)

its all good with [research.md](#) lets move ahead with the project

Agent

Thanks. Before we move on, I need two things settled. The instructions require your approval of the selected features to be **explicit**, and two source decisions are still open:

1. **Do you approve the five selected features** as drafted? (Image logging, agent/project estimator, digital day builder, device manufacturing, and like-for-like comparisons.)
2. **The two source decisions I left for you:** - **S15 (Altman)**: use as an unsupported company claim, or reject? - **S29 (Greenspector 2023)**: use with strong qualifications, or reject?

If you confirm, I'll record your approval and those two decisions in [research.md](#) exactly as you state them, and replace the "draft" markers with a note that you reviewed and approved it.

After that: - **The research transcript is a deliverable.** When you're ready, say **save transcript** and I'll save this whole conversation to `transcripts/research-YYYY-MM-DD_HHMMSS.md`. - **The next stage runs in a fresh**

chat. Start a new conversation and type **start specification**. I won't begin the specification in this chat.

38. Learner (RG Sureshkumar)

yes i approve of the features

Agent

[Tool: Bash — List draft status markers in research.md]

```
322:Status: draft, not yet reviewed by the user. Tags: P = improves representation of
professional AI use; D = connects AI use to broader digital life; X = cross-cutting candidate
for the strongest remaining need. No idea has been selected.
349:Status: draft. Decisions below are proposals for the user to confirm or change.
817:Status: draft of the user's selection, written for the user's review. N ... [truncated]
```

[Tool: Bash — Record user's review and approval in research.md status lines]

```
29:Researched activity by activity. Status: draft, not yet reviewed by the user.
29:Researched activity by activity. Reviewed by the user on 2026-09-26 ("its all good with
research.md").
322:Reviewed by the user on 2026-09-26. Tags: P = improves representation of professional AI
use; D = connects AI use to broader digital life; X = cross-cutting candidate for the
strongest remaining need. See Sele ... [truncated]
```

I've recorded your decisions in [research.md](#), using your own words:

- **Selected features:** marked as selected by you and **explicitly approved** on 2026-09-26 ("yes i approve of the features").
- **Audience needs, Possible features and Source assessments:** marked as reviewed by you on 2026-09-26 ("its all good with research.md"). Your approval covers my proposed source decisions.

Still open: you didn't say what to do with **S15 (Altman)** or **S29 (Greenspector 2023)**, so the file records both as still yours to decide. Neither is used by the five selected features' core data, so they don't block the specification. You can settle them any time.

The research transcript is a required deliverable. When you're ready, say **save transcript** and I'll save this whole conversation to transcripts/research-YYYY-MM-DD_HHMMSS.md, with user and agent turns marked and your name recorded as RG Sureshkumar.

After that, start a **fresh chat** and type **start specification** to begin the next stage.

39. Learner (RG Sureshkumar)

lets use S15

Agent

[Tool: Bash — Record the user's S15 decision in research.md]

349:Reviewed by the user on 2026-09-26, who accepted the decisions below except where marked as still open for the user (S29).
468:- Confidence and decision: low; use only as a stated company claim without a published method (decided by the user on 2026-09-26).

I've recorded your decision on S15 in [research.md](#). Altman's 0.34 Wh figure will be used only as a stated company claim with no published method, and the entry notes that you decided this on 2026-09-26.

Still open: S29 (Greenspector 2023). The choice is to use it with strong qualifications or reject it. It's only relevant to how social media appears in the digital day builder, so you can decide now or during the specification.

When you're ready, say **save transcript** and I'll save the research transcript.

40. Learner (RG Sureshkumar)

i want to reject S29

Agent

[Tool: Bash — Record the user's rejection of S29 in research.md]

349:Reviewed by the user on 2026-09-26, who accepted the decisions below; the user decided S15 and S29 directly.
468:- Confidence and decision: low; use only as a stated company claim without a published method (decided by the user on 2026-09-26).
580:- Confidence and decision: low; reject (decided by the user on 2026-09-26).
187:Evidence (see Source assessments S29-S32):
189:- Scrolling a feed fo ... [truncated]

Rejecting S29 also affects two other places in the document that cite it, so I'm updating those to match.

[Tool: Bash — Update references to S29 to reflect its rejection]

189:- Scrolling a feed for one minute on an Android phone over Wi-Fi was estimated at 0.47 gCO₂e (LinkedIn) to 0.96 gCO₂e (TikTok), with Instagram and YouTube at 0.87 and Facebook at 0.63 (S29, 2023; rejected by the user as a source, so not to be used for figures). A 2021 edition from the same company reportedly gave about 2.63 g per minute for TikTok and an average of about 1.15 g, but that page ... [truncated]

I've recorded your decision to reject S29 in [research.md](#), and updated the two places that cited it: - **Section 7** now says S29's per-minute figures aren't to be used. - **Selected features**, the digital day builder entry, now cites only S31, S32, S36 and S37 for weak categories, and notes that S29 and S30 were rejected.

What this means for the specification: with both Greenspector editions rejected and Greenly's figure disputed by TikTok, **no usable per-minute social media figure remains**. The evidence left is Meta's data-center share (S32) plus device-based streaming figures (S21, S25). So the modelling choice recorded in section 7 is effectively narrowed to treating social media as device use plus Meta's server share. That's for you to confirm when you write the specification.

All source decisions are now settled, and the research is approved by you.

When you're ready, say **save transcript** and I'll save the research transcript to transcripts/.

41. Learner (RG Sureshkumar)

save transcript

Agent

[Tool: Bash — Read transcript conventions, example, and PDF memory]

```
Exit code 1
# Transcripts

Session transcripts for this project, kept as evidence of process.

- `session-2026-09-12.md` — the build session: porting Andy Masley's calculator
  to a standalone site and adding the three features.
- `debrief-*.md` — Unit 1 debrief conversations, saved separately so debrief
  evidence stays distinct from build sessions.
- `debrief-*.pdf` — the same debrief rendered f ... [truncated]
```

[Tool: Bash — Read debrief transcript format, PDF memory, timestamp, session logs]

```
# Unit 1 Debrief — transcript

- **Project:** ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI
Explorations/Unit 1 Brief — AI Footprint Calculator
- **Repository:** https://github.com/rg-sureshkumar-goat/AI-Footprint-Calculator
- **Date:** 2026-09-11 → 2026-09-12
- **Learner:** RG Sureshkumar
- **Tool / model / effort (as reported by the learner):** Claude Code, Opus 5, ma ...
[truncated]
```

[Tool: Bash — Inspect session log structure and user turns]

```
1412
Counter({'assistant': 406, 'user': 253, 'attachment': 236, 'custom-title': 89, 'last-prompt':
88, 'agent-name': 88, 'atis-latch': 88, 'queue-operation': 82, 'file-history-snapshot': 41,
'system': 40, 'file-history-delta': 1})
1 None 'start research'
2 None 'lets go activity by activity'
```

```
3 None 'yes this list works, lets start with image generation'
4 None 'yes, add it to research.md'
5 None ' ... [truncated]
```

[Tool: Bash — Write transcript generator script to scratchpad]

```
d871894
https://github.com/rg-sureshkumar-goat/AI-Footprint-Calculator.git
ok
```

[Tool: Edit — ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator/transcripts/README.md]

```
The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI
Explorations/Unit 1 Brief - AI Footprint Calculator/transcripts/README.md has been updated
successfully.
```

Quick update: I've updated the transcripts README for the new research transcripts. Now I'm generating the transcript from this session's log, then I'll make the PDF.

[Tool: Bash — Generate sanitised research transcript and check redaction]